Subquadratic Cut-Query Algorithms for Directed Minimum Cuts*

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Abstract

We study exact minimum cuts in a simple unit-capacity digraph accessed only through queries giving the number of arcs leaving a vertex set. We give a deterministic minimum s–t cut algorithm using $O(n^{5/3}\log^2(2n))$ queries, a randomized terminal algorithm using $\widetilde{O}(n^{3/2})$ queries, and a randomized global algorithm using $\widetilde{O}(n^{7/4})$ queries. All return an original minimizing shore and its exact value; the randomized algorithms succeed with high probability.

Outgoing-cut queries cannot recover arc directions: reversing a directed cycle changes no cut value. However, the cut function depends only on how many arcs (zero, one, or two) join each pair of vertices and on each vertex's out-degree minus in-degree, its imbalance. The deterministic algorithm builds an integral digraph with the input's pair counts in which every pair it has not queried has a direction computable without queries. Its imbalances differ from the input's by only $O(n^{4/3})$ in total, and blocking flows correct them within the query budget; the corrected digraph has the input's cut function, so a maximum flow in it gives an exact minimum cut.

The randomized terminal algorithm instead optimizes over fractional orientations, which give each pair of vertices nonnegative real capacities in its two directions summing to the number of arcs joining it. It extends a linear program over fractional orientations from the earlier note *Directed Reachability from Cut Queries in $\widetilde{O}(n^{3/2})$ Queries*. We add a flow-value variable so that near-optimal solutions approximate the residual graph of a maximum flow; because cut values are integers, thresholding such a solution yields an exact minimum cut. We also give the numerical analysis needed to solve this linear program with cut queries. The global algorithm handles cuts with a small shore directly, and finds any other minimum cut by running the terminal algorithm on sampled pairs of terminals.

*This preliminary draft was written by Codex GPT-6 Astra; see the AI use statement just before the references. The main theorems have all been verified in Lean 4. Appendix A gives the correspondence between all numbered statements in the paper and their Lean counterparts, and notes where they differ. I have not yet reviewed the prose proofs here. Corrections are welcome.

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1 Introduction

An outgoing-cut query asks for the number of arcs leaving a specified set of vertices. Can these aggregate measurements find an exact minimum cut without revealing the underlying directed graph? This question separates two kinds of information that coincide in the usual adjacency model. Cut values suffice to specify the optimization problem, but they do not specify the arcs on which a flow algorithm would operate. Reversing a directed cycle, for example, preserves every outgoing-cut value.

We give subquadratic-query algorithms for exact terminal and global minimum cuts in simple unit-capacity digraphs. Our deterministic terminal algorithm uses $O(n^{5/3}\log^2(2n))$ queries. A different, randomized algorithm uses $\widetilde{O}(n^{3/2})$ queries for a terminal cut and yields a $\widetilde{O}(n^{7/4})$-query global algorithm. Every algorithm returns an actual minimum-cut shore of the input graph and its exact value.

The deterministic result rests on a representation theorem. We construct an implicit integral digraph with the same cut function as the input, whose undiscovered pairs have directions determined by known formulas. This representation supports residual search even though the original orientations remain unknown. A geometric bound on the divergence that must be corrected makes constructing it cheaper than learning all pairs. The randomized result takes another route: it optimizes a fractional representation, then extracts an exact minimum cut from a sufficiently accurate solution. Both routes use the oracle to operate on a representation suitable for optimization.

1.1 Model and results

The input is a loopless simple digraph $G = (V, E)$ on $V = [n]$, $n \geq 2$. Antiparallel arcs are allowed, with each ordered arc having capacity one. The only access to G is the exact oracle

$$F(S) = |E(S, V \setminus S)| \qquad (S \subseteq V).$$

For distinct terminals s, t, a terminal shore contains s and excludes t; a global shore is nonempty and proper. We write $\lambda_{st} = \min_{s\in S,\, t\notin S} F(S)$ and $\lambda = \min_{\varnothing\neq S\subsetneq V} F(S)$. Only calls to F are charged. Computation, storage, exact arithmetic, and enumeration between calls are unrestricted. Thus our bounds concern query complexity and do not assert polynomial running time. The notation $\widetilde{O}$ suppresses polylogarithmic factors in n and in the inverse failure probability when specified.

Problem	Algorithm	Outgoing-cut queries
Terminal minimum cut	Deterministic	$O(n^{5/3}\log^2(2n))$
Terminal minimum cut	Randomized	$\widetilde{O}(n^{3/2})$
Global minimum cut	Randomized	$\widetilde{O}(n^{7/4})$

All three bounds apply to the same input class and oracle, and include recovery of an exact minimizing shore.

Theorem 1.1 (Deterministic exact directed terminal cut)**.** *In the model above, a deterministic algorithm finds a minimum s–t shore and its exact outgoing-cut value using*

$$O(n^{5/3}\log^2(2n))$$

queries. Within the same bound, before choosing the terminal flow, it constructs an implicit integral directed representation having the same outgoing-cut function as G. Every unknown pair is pure in a public direction; every other pair has explicitly known capacities.

Here a pair is *pure* if all its capacity is in one direction, and *public* means computable from the transcript and stored data without an additional query. The representation can put capacity two in one direction, while preserving the sum of opposite capacities on each pair. Its maximum flow yields an original minimum-cut shore and value. It need not be a flow on the original hidden arcs, which the oracle cannot in general identify.

Theorem 1.2 (Randomized terminal and global cuts)**.** *In the same model, randomized algorithms find an exact minimum s–t cut in $\widetilde{O}(n^{3/2})$ queries and an exact global minimum cut in $\widetilde{O}(n^{7/4})$ queries, with probability at least $1 - n^{-c}$ for any fixed $c > 0$. Their query caps hold on failure executions as well. Every returned candidate is a legal original shore with its value checked by an outgoing-cut query.*

The deterministic construction and the randomized optimizer are separate algorithms. The former prepares an integral representation and runs blocking flows. The latter solves a linear program over fractional orientations and uses an integrality argument to recover a shore. The global algorithm balances a direct treatment of small shores against a small number of randomized terminal calls.

1.2 Context and the distinction from reachability

Rubinstein, Schramm, and Weinberg [RSW18] introduced cut queries to study exact optimization from aggregate graph information, motivated by value-oracle submodular function minimization. They gave randomized $\widetilde{O}(n^{5/3})$ terminal and $\widetilde{O}(n)$ global algorithms for unweighted undirected graphs, and asked whether weighted or directed cuts can be computed with $o(n^2)$ queries. For undirected graphs, later work gave $\widetilde{O}(n)$-query weighted global minimum cuts [MN20], $O(n)$-query global minimum cuts of simple graphs [AEG+22], and the cut-dimension approach to deterministic lower bounds [GPRW20, LLSZ21]. Anand, Saranurak, and Wang [ASW25] obtained deterministic $\widetilde{O}(n^{5/3})$ bounds using blocking flows. Jiang, Nanongkai, and Sawettamalya [JNS26] improved the randomized terminal bound to $\widetilde{O}(n^{8/5})$; Kenneth-Mordoch and Krauthgamer [KK26] subsequently obtained randomized all-pairs minimum cuts in $\widetilde{O}(n^{3/2})$ queries. These results exploit undirected structure.

A directed outgoing-cut function is submodular, so general submodular minimization finds a minimum terminal cut with $\widetilde{O}(n^2)$ value queries [LSW15, Jia21]. The best query lower bounds for submodular minimization are near-linear [GPRW20, CGJS22], while Graur et al. [GPRW20] also prove a $\binom{n}{2}$ query lower bound for finding a nontrivial minimizer of a general asymmetric submodular function. The open-problem discussion in [JNS26] identifies a polynomial saving over quadratic query complexity for directed terminal cuts as major progress toward subquadratic submodular minimization, and names directed reachability, raised by Nanongkai [Nan24], as a barrier already.

Directed reachability already illustrates the information obstacle. Cut queries determine a digraph only up to adding and deleting directed cycles [GPRW20], and Mon [Mon21] studies graph

problems in a directed cut-oracle model. Khanna, Putterman, and Song [KPS26] give $O(n \log^3 n)$-query reachability and topological sorting for directed acyclic graphs, and decide whether an arbitrary digraph has a cycle with the same number of queries. Bals, Tinca, and Nazari [BTN26] give a deterministic $O(n\sqrt{n \log n})$-query single-source reachability algorithm for directed acyclic graphs, and the same bound for topologically sorting the strongly connected components of a digraph whose component partition is known. Our input may contain arbitrary directed cycles, and we optimize cuts of positive value as well as detect zero cuts.

The closest prior work for our randomized algorithm is the anonymous note *Directed Reachability from Cut Queries in* $\widetilde{O}(n^{3/2})$ *Queries* [Anon26], listed in Yang P. Liu's repository on 12 August 2026 [Liu26]. It gives randomized reachability in general simple digraphs. Nanongkai's open-problem page now reports that several groups have reached roughly $n^{3/2}$ queries for reachability using ChatGPT [Nan24]. The note's central ingredients are a fractional-orientation LP of rank n, an implicit implementation of the Lee–Sidford framework [LS20], and filtered spectral sparsification. We use this viewpoint and spectral framework in the randomized algorithm. Our contribution is the extension to exact minimum-cut optimization, together with a replacement numerical execution that proves the needed implicit-LP guarantee. The deterministic representation theorem uses a different optimization problem and a different progress argument.

The distinction between reachability and exact optimization matters in the extraction step. The note's LP already penalizes the star variables that absorb divergence error, and thresholding its solution gives a digraph with the same zero out-cuts as the input [Anon26, Lemmas 3.3–3.4]. That suffices for reachability. For minimum cut, the optimum is unknown and may be positive. We add an explicit flow-value variable to the objective, so near-optimality simultaneously estimates the optimum and creates an approximate residual graph. Thresholding this graph, as in the note, produces a shore whose original cut value is less than one above optimum; integer input capacities then give exactness. This argument is described in Section 4.

1.3 Our contributions and the proof architecture

Three ideas account for the passage from cut measurements to exact directed optimization.

A query-efficient integral representation. The identity relating outgoing cuts to pair totals and vertex imbalances is already present in the reachability approach. The new deterministic step makes that identity algorithmically usable. Public offsets ensure that only a small set of pairs can remain fractional at any potential. Smoothing and integral rounding then leave a feasible divergence discrepancy. The decisive estimate bounds its total supply by $O(n^{4/3})$ through the height of the potential. Blocking flows [Din70], with the query accounting of [ASW25] that charges discoveries to path length, repair that discrepancy within the $\widetilde{O}(n^{5/3})$ budget.

Exact optimization from an implicit fractional residual. The randomized algorithm adds a flow-value variable to the penalized fractional-orientation LP of [Anon26], and turns an approximate LP solution into an exact minimum shore by one threshold search. Its cost is governed by the rank of the constraints, rather than the potentially quadratic number of hidden arc copies. We supply the regularized-weight and finite-precision analysis needed to execute the method with actual cut queries. Weighted path following and unregularized Lewis-weight estimates come from Lee and Sidford [LS20]; the graph oracle adapts weighted spanners and bundle sampling [BS07, Kou14], as proposed in [Anon26]. The exact-cut reduction and the repaired execution are proved here.

A global algorithm without recovering directions. For a threshold h, $O(nh \log n)$ queries learn enough pair information to handle every cut with a shore of size at most h. If an optimum has two larger shores, $\widetilde{O}(n/h)$ sampled terminal problems expose it. Balancing these costs with the terminal bound gives $\widetilde{O}(n^{7/4})$; this balance of small shores against sampled terminal pairs follows Cen, Li, Nanongkai, Panigrahi, and Saranurak [CLN+21]. The small-shore procedure evaluates directed cuts using known degrees and internal pair totals, so it needs no original arc orientations.

Both constructions distinguish an existence witness from information the algorithm must discover. An original flow proves that a repair demand is feasible or that the LP has a particular optimum, but it is never supplied to the algorithm. The proofs then account for every query needed to find the represented flow or its minimizing shore. This distinction is central in the cut-query model, where a short mathematical witness need not be cheap to discover.

Sections 2–5 explain these arguments and their query costs. Full proofs begin in Section 6. The deterministic and randomized query programs are formalized in Lean; Section 17 records the exact theorem statements and the polylogarithmic bounds covered by the accompanying formalization.

2 What the oracle reveals, and how to use it

For an unordered pair $e = \{u, v\}$ let c_e be the number of input arcs between its endpoints in both directions, so $c_e \in \{0, 1, 2\}$. Let $\beta_v = d^+(v) - d^-(v)$ be the vertex imbalance. Singleton and complementary-singleton queries learn both degree vectors in $O(n)$ calls. Summing imbalances over a shore cancels its internal arcs and gives

$$2F(S) = \sum_{e \in \delta(S)} c_e + \beta(S). \tag{1}$$

This is the cut algebra of the reachability note [Anon26, Lemma 2.2]. Thus any directed representation with these pair totals and imbalances has exactly the input's cut function. The two cyclic orientations of a triangle illustrate both the ambiguity and its resolution: their arc directions differ, but both have pair totals one and imbalance zero. Either orientation gives the same optimization problem.

Learning all c_e by individual pair tests would take a quadratic number of queries, and general graph reconstruction from additive queries saves only a logarithmic factor [GK00]. Instead we use partial stars, as in [Anon26, Corollary 2.3]. For $v \notin T$, cancellation gives

$$\sum_{u \in T} c_{uv} = F(\{v\}) + F(T) - F(T \cup \{v\}). \tag{2}$$

The right-hand side is nonnegative and vanishes exactly when no positive pair joins v to T. Bisection therefore discovers a positive pair in $O(\log n)$ queries. The set T may be specified by any known predicate on vertex labels and the transcript. In particular, it can select pairs with a given direction in an implicit representation, or whose publicly computed coefficient exceeds a threshold. Such a predicate is defined on all potential pairs, including nonedges.

The same identity evaluates weighted aggregates, by the bit-class aggregation of [Anon26, Lemma 2.4]. Suppose a signed coefficient a_{uv} is publicly computable and has B binary positions. For each vertex, sign, and position, query the set of endpoints with that bit present. Summing the resulting star counts computes $\sum_u c_{uv} a_{uv}$ for every v in $O(nB)$ queries. This implements the gradients and matrix products used below. The coefficients are chosen by the algorithm; the unknown quantities remain the bounded integers c_{uv}. No weighted-cut oracle is assumed.

To express orientations, fix a canonical direction on each pair and let d_e be its incidence column, with $+1$ at the tail. Write D for the matrix of all such columns. A signed value $z_e \in [-c_e, c_e]$ represents capacities $(c_e + z_e)/2$ and $(c_e - z_e)/2$ in the canonical and reverse directions. Its imbalance is Dz. The original graph proves that $Dz = \beta$ is feasible, although its corresponding z is unknown.

An arbitrary feasible orientation is still insufficient for a cheap flow algorithm: searching its outgoing arcs might require inspecting all pairs. We therefore seek additional structure. The deterministic algorithm makes every unknown pair pure in a public direction. The randomized algorithm maintains a public fractional formula on each pair and searches only after thresholding it. In each case the star identity turns the representation's search predicate into legal original queries. This interface is the common foundation; the constructions that create the two useful representations are different.

3 The deterministic construction: a representation we can repair

Theorem 1.1 is proved by preparing an integral representation before solving the requested terminal problem. We first arrange that every unknown pair has a public direction, while tolerating an imbalance b different from β. The remaining task is a transshipment: route the integral demand

$$q = (b - \beta)/2, \qquad R = \sum_v q_v^+.$$

Replacing the used capacities by residual capacities changes the imbalance from b to $b - 2q = \beta$, and preserves every pair total. By (1), the repaired representation is cut-equivalent to G. The factor two arises because sending flow decreases a forward residual capacity and increases the reverse capacity by the same amount.

There are two costs to control: obtaining the public directions, and repairing their imbalance. Our construction uses a potential $r \in \mathbb{R}^V$ and a parameter k. It achieves the following three properties:

$$\text{at most } O(n^2/k + n\log(2nk)) \text{ pairs need explicit learning,} \tag{3}$$

$$b - \beta = -r + e, \qquad \|e\|_1 = O(n\sqrt{k}), \qquad \sum_v r_v = 0, \tag{4}$$

$$r_w - r_v \le 2 \quad \text{on every positive represented arc } v \to w. \tag{5}$$

The first property makes the representation searchable. The second relates the repair demand to a potential. The third turns high potential into long flow paths, giving a stronger repair bound than a norm estimate alone. We explain the construction and the two flow estimates in turn.

3.1 Sparse bands that work for every potential

Orienting each pair by a potential difference is a natural way to obtain public directions. Its difficulty is ties: a large number of pairs can have nearly equal endpoint potentials, and making all of them explicit would exhaust the budget. We separate their switching thresholds by assigning public offsets $\omega_e \in [-1, 1]$ to all potential pairs. For a scale $1/k$, define the band

$$\mathcal{A}(r) = \{e : |d_e^\top r - \omega_e| < 1/k\}.$$

Lemma 7.1 constructs one offset table satisfying (3) for *every* real vector r. The table is chosen before querying the input. Uniformity over all potentials is essential: the eventual r depends on the graph and on earlier answers.

The existence argument is a finite discretization. Give each offset an independent uniform value in $\{-k, \dots, k\}/k$. For a fixed potential, each pair lies in the band with probability $O(1/k)$. To cover all potentials, observe that an active edge has endpoint difference at most two. Translating the potentials independently on each component of the active graph puts all coordinates in $[0, 2n]$, without losing an active edge. Rounding to a grid of spacing $1/(4k)$ enlarges the band by only a constant factor. There are $(O(nk))^n$ grid vectors, so concentration and a union bound add $O(n\log(nk))$ to the expected band size $O(n^2/k)$.

Randomness here proves the existence of a fixed finite table. Enumerating tables and testing the relevant rational difference constraints selects one deterministically, with no input queries. This is one place where unrestricted offline computation is used. Once fixed, the table permits the algorithm to learn every exceptional pair even before knowing which of those pairs are edges.

3.2 Smoothing finds directions and exposes the repair demand

The offsets are incorporated into an objective whose gradient is available through the aggregate oracle. Let ϕ_δ be the Huber smoothing [Hub64] of absolute value: it equals $x^2/(2\delta)$ for $|x| \le \delta$ and $|x| - \delta/2$ otherwise. With $\delta = 1/k$, minimize

$$\mathcal{F}(r) = \sum_e c_e \phi_{1/k}(d_e^\top r - \omega_e) - \beta^\top r + \tfrac{1}{2}\|r\|_2^2. \tag{6}$$

The quadratic makes $\mathcal{F}$ strongly convex. Its gradient is

$$\nabla \mathcal{F}(r) = D\big(c_e \operatorname{clip}(k(d_e^\top r - \omega_e), -1, 1)\big) - \beta + r.$$

For a given r, the clipped coefficients are known functions on pairs. Rounding them to a fixed $O(\log n)$-bit denominator evaluates an approximate gradient in $O(n \log n)$ cut queries. The symmetrized graph has maximum weighted degree at most $2(n-1)$, so the objective is $O(nk)$-smooth. Nesterov's accelerated iteration [Nes83, Nes04], with controlled absolute gradient error, uses $O(\sqrt{nk}\log n)$ steps. The total is $O(n\sqrt{nk}\log^2 n)$ queries. The algorithm never evaluates the unknown objective value or needs to know its minimizer.

Interpret the clipped coefficients as a fractional orientation. A coefficient $+1$ or -1 gives a pure pair in a public direction. Every coefficient strictly between them belongs to $\mathcal{A}(r)$. Learn these pairs, then round their capacities integrally by the usual circulation rounding argument. Each vertex's signed divergence changes by at most two. The approximate stationarity equation then gives (4). Saturation of the clipped coefficients ensures (5): before rounding, a positive arc cannot climb by more than $1 + 1/k \le 2$, and rounding only affects pairs already within the narrow band. The details are in Section 7.

The resulting discrepancy is feasible, even though we have never learned the original orientation. On a canonical pair, let a'_e be the rounded forward capacity and a_e^G the unknown original forward capacity. The signed amount $a'_e - a_e^G$ fits in the rounded representation in its appropriate direction. Its divergence is exactly $(b-\beta)/2$. Moreover, b_v and β_v have the same parity, because both arise from integral orientations of the same pair totals. Hence q is an integral, zero-sum, feasible transshipment. The original graph is used only to prove feasibility; the repair algorithm receives the explicit vector q and the searchable rounded representation.

3.3 Potential height bounds the amount that must be repaired

The key estimate is

$$R \le 4n^{4/3} + \tfrac{5}{4}\|e\|_1. \tag{7}$$

It follows from feasibility and (5), not from further queries. A generic Euclidean estimate on the potential would only give $R = O(n^{3/2})$, leading to a larger repair cost. The height argument extracts information from the geometry of a feasible flow.

Ignore the error e temporarily. Vertices with negative potential supply $-r_v/2$ units, while those with positive potential demand $r_v/2$. A path serving a vertex at height at least h must climb from negative potential to height h. Since each arc climbs by at most two, every such path has length $\Omega(h)$. Consider the network restricted only at its sources and destinations to this part of a feasible transshipment; keep all represented internal arcs. Its shortest terminal distance is still $\Omega(h)$.

Among the middle BFS layers of a graph at terminal distance d, there is a consecutive pair with only $O(n/d)$ vertices in total; this is the classical layer argument of Karzanov [Kar73] and Even and Tarjan [ET75]. The capacity of the associated prefix cut is $O(n^2/d^2)$, because opposite capacities on each internal pair sum to at most two. Thus at most $O(n^2/h^2)$ units can travel from negative potential to destinations at height h. The auxiliary source and sink capacities may be large; they do not cross these middle prefix cuts.

Restoring e charges misclassified supplies and perturbed demands to $O(\|e\|_1)$. The resulting inequality is

$$\sum_{v:r_v \ge h} r_v \le 2\|e\|_1 + \frac{64n^2}{h^2}.$$

The remaining positive potentials sum to at most nh. Since r has zero sum and $q = (-r + e)/2$, we obtain

$$R \le \frac{nh}{2} + \frac{5}{4}\|e\|_1 + \frac{32n^2}{h^2}.$$

Choosing $h = 4n^{1/3}$ proves (7). The entire reasoning concerns an analysis flow whose existence is known; discovering it is unnecessary. Nor must the potential-rise property survive the later repair. It is needed only to bound the initial demand.

3.4 Charging flow discovery to path length

We now route q. Add an explicit supersource and supersink with capacities q_v^+ and q_v^-, and run Dinic's integral blocking-flow phases [Din70]. Undiscovered pairs have never carried flow, so their residual direction is still their public direction. The star oracle discovers an eligible positive pair when needed; afterward both residual capacities are stored explicitly. Known arcs, including the possibly heavy terminal arcs, require no further queries.

As in Anand, Saranurak, and Wang [ASW25], the useful accounting is by the work of a whole blocking phase. If its residual distance is d and its augmentation value is a, the cost is

$$O((n + da)\log n). \tag{8}$$

BFS costs $O(n \log n)$ queries. During depth-first search in the frozen level graph, a failed new traversal makes its destination permanently dead for the phase. There are at most n such failures. A successful new traversal is used in the next complete augmenting path. Integral augmentation puts at least one unit on that discovered pair. All phase flow goes forward in the levels, so its total arc mass is da and these charges cannot be canceled within the phase. This bounds discoveries even when scanning many already known arcs costs substantial offline time.

The middle-layer argument also bounds the remaining deficit at distance d by $O(n^2/d^2)$ [Kar73, ET75]. Distances strictly increase between phases. If the phases have distances d_i and values a_i, then

$$\sum_i d_i a_i = \sum_{j \geq 1} \sum_{i : d_i \geq j} a_i = O(n\sqrt{R}).$$

To see the last bound, use R for the tail mass at small j, use $O(n^2/j^2)$ at large j, and split at $j \asymp n/\sqrt{R}$. After $O(n^{2/3})$ phases, only $O(n^{2/3})$ integral units can remain, so the number of phases is also $O(n^{2/3})$. Summing (8) gives

$$O((n\sqrt{R} + n^{5/3}) \log n) \tag{9}$$

queries, including all auxiliary terminal searches. This is Theorem 9.1.

3.5 Where the exponent comes from

Set $k = \lceil n^{1/3} \rceil$. The smoothed optimization and exceptional pair learning cost

$$O\left(n\sqrt{nk}\log^2 n + \frac{n^2}{k} + n\log(nk)\right) = O(n^{5/3}\log^2 n).$$

Equations (4) and (7) give $R = O(n^{4/3} + n\sqrt{k}) = O(n^{4/3})$, so the repair fits within the same budget. The residual representation now has imbalance β and is cut-equivalent to G. A final blocking-flow solve for s, t costs $O(n^{5/3}\log n)$: its total flow is at most $F(\{s\}) \leq n - 1$, and the same discovery analysis applies. By the max-flow min-cut theorem [FF56], the final residual reachable set is a minimum shore of both graphs.

Operation	Queries at the chosen parameters
Evaluate accelerated gradients	$O(n^{5/3}\log^2 n)$
Learn all potentially fractional pairs	$O(n^{5/3})$
Repair the divergence discrepancy	$O(n^{5/3}\log n)$
Solve the requested terminal problem	$O(n^{5/3}\log n)$

There are two separate balances behind this accounting. The algorithmic parameter k balances the optimization cost $n^{3/2}\sqrt{k}$ against the learning cost n^2/k. The analysis threshold h balances nh against n^2/h^2 in the height estimate. Both scales are of order $n^{1/3}$, but only k is used to construct the representation. Stopping at the coarser bound $R = O(n^{3/2})$ would leave a repair cost $n\sqrt{R} = O(n^{7/4})$. The height estimate is what makes repair as cheap, up to logarithms, as preparing the public directions.

Bounded pair capacity controls two parts of the argument. It bounds the height tail before repair and the remaining flow deficit during repair. Integrality enables the discovery charges and bounds the number of late phases. These are the reasons for the simple unit-capacity input assumption, and for keeping that assumption explicit in the theorem.

4 The randomized algorithm: optimize a fractional residual

The randomized algorithm avoids building an integral orientation. It searches directly for a fractional representation resembling the residual network of a maximum flow. Two ingredients are needed: an LP whose near-optimal solutions encode such residuals, and a way to solve it without materializing its hidden coordinates. The first ingredient provides the exact-mincut extension of the reachability framework. The second uses rank-sensitive path following, with the graph oracle and numerical guarantees made explicit.

4.1 A penalty that controls both value and divergence

Fix $s \neq t$ and write $\chi = e_s - e_t$ and $\lambda = \lambda_{st}$. Conceptually replace each pair e by c_e identical copies. Let B have one canonical incidence row per copy and introduce a variable $z_{e,j} \in [-1, 1]$ for its fractional orientation. Thus the copy has capacities $(1 + z_{e,j})/2$ and $(1 - z_{e,j})/2$ in its two directions. Add an explicit variable q for twice the proposed flow value. As in the reachability LP of [Anon26, eq. (4)], nonnegative star variables y^+, y^- absorb divergence error and are penalized in the objective. The LP is

$$\begin{aligned} \text{minimize} \quad & R - q/2, && R = \sum_v (y_v^+ + y_v^-), \\ \text{subject to} \quad & B^\top z + q\chi + y^+ - y^- = \beta, \\ & -1 \le z_{e,j} \le 1, \quad -2 \le q \le 2n, \quad 0 \le y_v^+, y_v^- \le 4n. \end{aligned} \tag{10}$$

All copies of a pair have identical data. The algorithm maintains one public formula for their common coordinate, so it need not know which copies exist. The star variables supply a known strictly feasible start: $z = q = 0$ and $y_v^\pm = 2n \pm \beta_v/2$.

Let $f = q/2$, $\rho = y^+ - y^-$, and let P_z be the represented fractional digraph. Its imbalance is $\beta - 2f\chi - \rho$ and its pair totals are those of G. The cut identity gives

$$P_z(S) = F(S) - f\chi(S) - \rho(S)/2. \tag{11}$$

For a minimum terminal shore S_*, nonnegativity implies $f \le \lambda + R/2$, so $R - f \ge -\lambda + R/2$. Conversely, the actual residual network of a maximum flow in G yields a feasible point with $f = \lambda$ and $R = 0$. Therefore the LP optimum is $-\lambda$, and every feasible ϵ-suboptimal solution satisfies

$$R \le 2\epsilon, \qquad |f - \lambda| \le \epsilon. \tag{12}$$

The penalty has two simultaneous effects: it recovers the optimum value and makes the represented imbalance close to that of a maximum-flow residual. Merely finding an approximately feasible orientation would not give this conclusion. The proof uses an original maximum flow as an existence witness, without querying or recovering that flow.

4.2 One threshold search recovers an exact minimum shore

An approximate objective value alone does not identify a minimizing set. Equation (11) provides the additional structure. Choose a dyadic threshold $\tau \le 1/(8n^2)$ of order n^{-2} and solve the LP to error $\epsilon = \tau/16$. For each ordered potential pair, evaluate its per-copy fractional capacity p_{uv} to additive error at most $\tau/4$, obtaining a public value $\widehat{p}_{uv}$. As in the threshold digraph of [Anon26, Lemmas 3.3–3.4], keep an arc in an implicit graph J exactly when

$$c_{uv} > 0 \quad \text{and} \quad \widehat{p}_{uv} > \tau.$$

Find the vertices W reachable from s using filtered star searches.

Optimality follows from two bounds. On a true minimum shore, (12) implies $P_z(S_*) \le 2\epsilon = \tau/8$. No individual outgoing fractional arc can exceed this total, even before rounding. Hence no arc of J leaves S_*, and the reachable set satisfies $s \in W \subseteq S_*$ and $t \notin W$. In the other direction, no retained arc leaves W, so every outgoing per-copy capacity of W is at most $5\tau/4$. There are fewer than n^2 copies. Consequently

$$F(W) = P_z(W) + f + \rho(W)/2 < \lambda + 2\epsilon + \tfrac{5}{4}n^2\tau < \lambda + 1.$$

Since $F(W)$ is an integer and W is a legal terminal shore, $F(W) = \lambda$. The rounded threshold is separated from both relevant bounds, so the argument does not require exact comparison with an irrational fractional coordinate.

Each successful search discovers a new vertex, and each scanned vertex has at most one final unsuccessful query sequence. Thus extraction uses only $O(n \log n)$ queries. One final query returns the exact value. On a failed solver execution the procedure still returns a legal checked shore, using $\{s\}$ if necessary. This permits amplification by taking the least checked value; an incorrect candidate cannot have value below the true optimum. Section 10 and the two following sections give the LP reduction and extraction proof in full.

4.3 How a quadratic-size LP costs only $\widetilde{O}(n^{3/2})$ queries

The constraint matrix has n columns and up to $\Theta(n^2)$ hidden incidence rows, together with $2n + 1$ explicit rows. Its rank is n. The rank-sensitive Lee–Sidford method [LS20] suggests $\widetilde{O}(\sqrt{n})$ optimization batches. To turn this into a query bound, each batch must be executable without reading its hidden rows. The fractional-orientation framework of [Anon26] makes this possible by retaining common public formulas on duplicate copies.

Normal matrices in the execution have the form

$$E + \sum_e c_e w_e d_e d_e^\top, \tag{13}$$

where E is explicit and positive semidefinite, while w_e is a public coefficient computed from the current iterate. A spectral approximation of the Laplacian sum can be learned in $\widetilde{O}(n)$ queries. Adding E preserves the approximation. Solves in this explicit approximation and all other offline linear algebra cost no queries. Products and residual corrections are computed by signed binary aggregates as in Section 2.

The spectral construction follows the filtered-spanner approach of [Anon26], adapting weighted spanners [BS07] and bundle sampling [Kou14]. Assign a potential pair length $1/w_e$ and sort eligible endpoints by their public lengths. Binary star search finds the first present pair in such an ordering, without inspecting all shorter nonedges. This implements the required minimum-edge searches for a weighted spanner. A bundle of edge-disjoint spanners gives many short paths for each remaining pair, bounding its effective resistance and hence the sampling probability needed to preserve the Laplacian [SS11, KP12]. Recursive sampling produces an explicit spectral approximation.

Three features of this interface matter to the LP. Searches use public pair masks, so they remain valid when the weights depend on earlier answers. Sampling treats the entire pair together, retaining the common formula on its copies. Finally, the query count is independent of the ratio between the largest and smallest positive public weights: the algorithm compares known lengths instead of searching weight scales. Fresh coins and fixed caps give a guarantee conditional on every earlier transcript. Section 14 proves this interface.

4.4 The additional analysis needed for an executable LP bound

A fast spectral oracle is only one component of an implicit optimizer. The execution must also control errors in public coordinate formulas, normal solves, and weight updates, while maintaining feasibility and staying inside the variable boxes. Our analysis supplies these guarantees for the terminal LP. It replaces the inexact-solver interface invoked in the earlier reachability note; the exact-cut reduction does not assume that interface as a black box.

One issue is regularization of Lewis weights, which measure the influence of the rows in the changing normal systems. Their unregularized sum is the constraint rank. A positive regularizer bounds every weight away from zero, but a uniform sensitivity claim requires control of its scale. We use $g_i = w_i + c_0$ with $c_0 = 1/(2\overline{M})$ and $M \le \overline{M} < 2M$, where the row count $M = |E| + 2n + 1$ is known from the degree queries. A light/heavy row decomposition proves sensitivity at most 16. The light rows have bounded total normalized mass; on the heavy rows the regularizer changes the weights by at most a constant factor. This is the content of Lemma 15.2.

The second issue is error propagation through feasible Newton steps. A Pythagorean estimate controls projection in the mixed norm used by the path-following method. Normal solves and scalar evaluations use absolute error budgets, and an explicit star right inverse corrects each primal step to exact feasibility. A polynomial lower bound on the slack then justifies fixed inverse-polynomial precision throughout the execution. These estimates are proved together, so the precision requirements follow from the iterates that the algorithm actually produces.

The feasibility correction has a direct oracle implementation. If a rounded update has constraint residual $v = A^\top x - b$, signed aggregates measure v in $\widetilde{O}(n)$ queries. The explicit $+I_n$ rows allow a correction $-v$ in their coordinates, restoring the equality exactly. The slack bound and absolute error budget ensure that this correction stays within the boxes. Thus approximate linear algebra cannot accumulate an uncontrolled divergence error in the fractional residual used for cut extraction.

The resulting implementation performs $\widetilde{O}(\sqrt{n})$ batches, with $\widetilde{O}(n)$ queries per batch, and returns an exactly feasible, inverse-polynomially accurate solution whose hidden coordinates have public formulas. Combining it with the threshold search proves the randomized terminal bound. Section 15 contains the full sensitivity, projection, slack, and finite-precision arguments. The contribution here is this complete execution and its exact-mincut application; rank-sensitive path following and the spectral construction's underlying spanner method remain attributed to their sources.

5 From terminal cuts to a global minimum

A fixed root and every other vertex would reduce global mincut to $2(n-1)$ directed terminal calls. To use fewer calls, we handle all small shores together and sample terminals only for cuts whose two shores are large, following the balance used by Cen, Li, Nanongkai, Panigrahi, and Saranurak [CLN+21] for directed minimum cuts. The essential local step learns internal pair totals, which are enough to evaluate the relevant directed cuts even though the original orientations remain hidden.

Let $\delta = \min_v\{d^+(v), d^-(v)\}$; a known singleton or complementary-singleton shore attains this value. Fix $1 \le h < n$. We obtain, in $O(nh \log n)$ queries, a candidate at least as good as every cut with a shore of size at most h. If $\delta > h$, the degree candidate already suffices: a shore of size $s \le h$ has at least

$$s\delta - s(s-1) = \delta + (s-1)(\delta - s) \ge \delta$$

outgoing arcs, with the symmetric indegree argument for a small complement. This is where simplicity controls the number of internal arcs.

Suppose instead that $\delta \le h$. Set

$$D_+ = \{v : d^+(v) \le \delta + h\}, \qquad D_- = \{v : d^-(v) \le \delta + h\}.$$

The induced symmetrizations on these sets have at most $2nh$ copies each, by summing outdegrees or indegrees. Learn them by star searches. For any $S \subseteq D_+$ we can now evaluate, with no further queries,

$$F(S) = \sum_{v \in S} d^+(v) - \sum_{e \subseteq S} c_e.$$

The analogous indegree identity evaluates $F(V \setminus T)$ for $T \subseteq D_-$. Any shore S of size at most h with $F(S) < \delta$ lies entirely in D_+, since each of its vertices has $d^+(v) \le |S| - 1 + F(S) < h + \delta$. A small complement lies in D_-. Enumerating the relevant subsets offline therefore finds the required candidate. This argument explains both the query saving and the role of unrestricted offline computation.

Fix a root a and sample $O((n/h)\log n)$ vertices. For each sampled $v \ne a$, solve both ordered terminal problems (a, v) and (v, a). If a fixed global optimum has two shores larger than h, a sample lands opposite a with probability at least h/n. With high probability one of these calls is separated in the direction of that optimum, and its minimum value equals the global minimum. Taking the least checked candidate handles both cases, at cost

$$\widetilde{O}\left(nh + \frac{n}{h} Q_{st}(n)\right). \tag{14}$$

With $Q_{st}(n) = \widetilde{O}(n^{3/2})$, the balance $h \asymp n^{3/4}$ gives $\widetilde{O}(n^{7/4})$. Every terminal call uses the original graph; the reduction needs neither a contraction oracle nor directed arc recovery.

The full reduction is Theorem 10.4. Together with the two terminal constructions, it establishes all three bounds stated at the outset. Sections 6–17 provide the detailed oracle implementations, numerical analysis, and formalization statements.

6 The common oracle representation

Write $a^G_{uv} = \mathbf{1}[(u,v) \in E]$. Query the singleton and complementary-singleton shores to obtain

$$d^+(v) = F(\{v\}), \quad d^-(v) = F(V \setminus \{v\}), \quad \beta_v = d^+(v) - d^-(v).$$

For an unordered pair $e = \{u, v\}$ put $c_e = a^G_{uv} + a^G_{vu} \in \{0, 1, 2\}$. Let U be the undirected graph of pair capacities c_e and let $c_U(\delta S)$ denote its cut capacity. For each potential pair $u < v$, use incidence column $d_e = e_u - e_v$; D is the matrix of all these columns, including those of zero pairs. All sums involving c_e may therefore be indexed by the public universe $\binom{V}{2}$.

The identities below are the cut algebra (Lemma 2.2) and the filtered-neighbor oracle (Corollary 2.3) of the reachability note [Anon26]; we include proofs for completeness.

Lemma 6.1 (Cut algebra and star access)**.** *For every shore S and every $v \notin T$,*

$$2F(S) = c_U(\delta S) + \beta(S), \tag{15}$$

$$c(v,T) := \sum_{u \in T} c_{uv} = F(\{v\}) + F(T) - F(T \cup \{v\}), \tag{16}$$

$$F(S) = \sum_{v \in S} d^+(v) - \sum_{e \subseteq S} c_e. \tag{17}$$

A directed capacitated representation with pair totals c_e and divergence β has the same outgoing-cut function as G.

Proof. Let $I(S)$ be the incoming cut. Summing divergences over S gives $\beta(S) = F(S) - I(S)$, while $c_U(\delta S) = F(S) + I(S)$. This proves the first identity and cut equivalence. For disjoint sets A, B, the arcs canceled in $F(A) + F(B) - F(A \cup B)$ are precisely the arcs between A and B in either direction. Set $A = \{v\}$ for the second identity. Finally, summing the outdegrees in S counts its outgoing cut and each internal directed arc once. The internal sum is $\sum_{e \subseteq S} c_e$. □

Corollary 6.2 (What the outgoing-cut function determines)**.** *Two loopless capacitated digraphs on the same labeled vertex set have the same outgoing-cut function if and only if their symmetrized pair capacities and vertex imbalances agree.*

Proof. The cut function determines $\beta_v = F(\{v\}) - F(V \setminus \{v\})$ and $c_{uv} = F(\{u\}) + F(\{v\}) - F(\{u,v\})$. Conversely these quantities determine every cut by (15). □

Graur et al. [GPRW20] observed that cut queries determine a weighted digraph only up to adding and deleting directed cycles, and Ghosh et al. [GKMM26] that the skeleton and indegrees of a digraph determine every property invariant under cycle reversal; the corollary is the corresponding statement for outgoing cuts. The statement is about information, not the cost of learning all pairs. For example, the two cyclic orientations of a triangle have $c_{uv} = 1$ on each pair and $\beta = 0$, so no outgoing-cut query distinguishes them. The algorithms below retain most pair capacities implicitly and expose only those needed for optimization and residual search.

A formula or predicate on potential pairs is *public* if it is computable from vertex labels, the transcript, explicitly stored data, and public coins, without another adjacency lookup. It must also be defined on nonedges. Thus a query may use $T = \{u : P(v,u)\}$ for a public predicate P, although the algorithm does not know which of these pairs are present. These are the pair-evaluable quantities of [Anon26, Definition 2.1].

The next lemma restates the filtered-neighbor search and bit-class aggregation of [Anon26, Corollary 2.3 and Lemma 2.4].

Lemma 6.3 (Filtered discovery and exact aggregates)**.** *A positive pair incident to v in a public set T can be found, or its absence certified, in $O(\log(2n))$ outgoing queries. The first such pair in any public ordering can be found in the same bound. Its multiplicity is learned exactly. If a public signed dyadic formula $q(v,u)$ uses at most B fixed binary positions, all values*

$$\sum_{u \neq v} c_{uv} q(v,u) \qquad (v \in V)$$

can be computed exactly in $O(nB)$ queries. So can $\sum_{u<v} c_{uv} q(u,v)$.

Proof. A star count is nonnegative and is zero precisely when there is no positive pair. Bisect the public list, retaining the first half if its count is positive and the second half otherwise. A final singleton count reveals the pair capacity. After the singleton setup, a star count needs at most two new calls to F.

For aggregation, fix a vertex, sign, and bit position. Form the public set of endpoints whose coefficient has that sign and that bit in its magnitude. Its star count, multiplied by the signed power of two, is the corresponding contribution. Summing over the B positions and two signs recovers the row. For a scalar pair sum restrict row u to $v > u$. This avoids double counting. In particular, public integers $h_e \in [-Q, Q]$ give an exact measurement of $D(c_e h_e/Q)$ in $O(n \log(2Q))$ queries. □

7 A public integral representation with small discrepancy

We give the deterministic construction underlying Theorem 1.1. It uses the potential pair incidence matrix D and multiplicities c_e of Section 6. The original signed pair capacities show that

$$\beta \in \{Dz : |z_e| \leq c_e\}. \tag{18}$$

We shall construct integral capacities on each pair whose sum is c_e. Every unknown pair will be pure in a public direction; all split pairs will have been learned. Its divergence, denoted b, is measured exactly. The repair demand is $q = (b - \beta)/2$. Constructing a representation with small $\|b - \beta\|_1$ would already be useful, but the potential structure of q will give a stronger bound than a Euclidean discrepancy estimate alone.

7.1 One offset table works for all potentials

Lemma 7.1 (Uniform sparse bands)**.** *For every positive integer k, there are public offsets $\omega_e \in \{-k, \ldots, k\}/k$ such that, for every $r \in \mathbb{R}^n$, the set*

$$\mathcal{A}(r) = \{e : |d_e^\top r - \omega_e| < 1/k\}$$

has size at most

$$H = \left\lceil \frac{6\binom{n}{2}}{k} \right\rceil + 2n\lceil \log_2(8nk+1) \rceil + 1 = O(n^2/k + n\log(2nk)). \tag{19}$$

The table is deterministically constructible without an input query.

Proof. Use independent uniform offsets on the stated grid only to prove existence. If a pair is active then $|r_i - r_j| < 1 + 1/k \leq 2$. Translate the potentials on each component of the active graph until its minimum is zero. Every coordinate is then in $[0, 2n]$: connect a vertex to the minimum by a simple path of at most $n - 1$ active edges. All formerly active edges stay active. Round

each coordinate down to a multiple of $1/(4k)$. Every formerly active edge now satisfies the wider inequality $|r_i - r_j - \omega_e| < 2/k$.

There are at most $(8nk+1)^n$ such grid vectors. For any fixed one, a particular offset lies in its closed band of radius $2/k$ with probability at most $5/(2k+1) \le 3/k$. If X counts these pairs, independence gives

$$\mathbb{E}e^X \le \exp\left(\frac{6\binom{n}{2}}{k}\right),$$

since $e - 1 < 2$. Markov's inequality and the union bound show that the probability of some grid vector having more than H pairs in its wider band is less than one. Consequently some offset table satisfies the original narrow-band bound for every real potential.

Enumerate all finite offset tables in a fixed order. For each, test whether some set of $H+1$ potential pairs admits the strict rational difference constraints $|r_i - r_j - \omega_e| < 1/k$. This is a finite enumeration followed by rational linear feasibility with a positive common slack. Components may be translated into $[0, 2n]$ as above, so a bounded rational LP suffices. If $H \ge \binom{n}{2}$, accept immediately. The first table passing all tests exists. No test uses the unknown c_e, and one accepted table works for all adaptively chosen potentials later in the algorithm. □

7.2 Huber smoothing and cut-implemented gradients

Fix this table and put $\delta = 1/k$. Define the Huber function [Hub64] ϕ_δ and the objective f by

$$\phi_\delta(z) = \begin{cases} z^2/(2\delta), & |z| \le \delta, \\ |z| - \delta/2, & |z| > \delta, \end{cases} \qquad f(r) = \sum_e c_e \phi_\delta(d_e^\top r - \omega_e) - \beta^\top r,$$

and let $\mathcal{F}(r) = f(r) + \|r\|_2^2/2$. The regularization parameter is fixed at one. The objective is used for the analysis; its value and its unknown minimum are not queried.

Lemma 7.2 (Geometry of the regularized objective)**.** *The function $\mathcal{F}$ is 1-strongly convex and $(4nk+1)$-smooth, with*

$$\nabla f(r) = D(c_e \operatorname{clip}(k(d_e^\top r - \omega_e), -1, 1)) - \beta. \tag{20}$$

Writing $m = \sum_e c_e \le n(n-1)$ and r_* for its unique minimizer,

$$f(r) \ge -3m/2, \quad f(0) \le m, \quad \mathcal{F}(0) - \mathcal{F}(r_*) \le 5n^2/2, \quad \|r_*\|_2^2/2 \le 5n^2/2. \tag{21}$$

Proof. Differentiation gives (20); the clipped derivative has slope at most k. The c-Laplacian has largest eigenvalue at most $2\max_v \sum_{u \ne v} c_{uv} \le 4(n-1)$, so f is $4nk$-smooth. Adding the quadratic proves strong convexity and smoothness. By (18), $\beta^\top r \le \sum_e c_e |d_e^\top r|$. Use $\phi_\delta(z) \ge |z| - \delta/2$ and $|\omega_e| \le 1$ to get $f(r) \ge -(1 + \delta/2)m$. Also $\phi_\delta(-\omega_e) \le |\omega_e| \le 1$. These two bounds prove both estimates involving r_*, using $\mathcal{F}(r_*) \le \mathcal{F}(0)$. Coercivity and strict convexity give existence and uniqueness. □

Let T be the least integer with $T^2 \ge 8nk$, set $L = T^2$ and $\alpha = 1/T$, and let Q be the least power of two at least $64n^2$. For any public rational v, round each clipped coefficient in (20) down to h_e/Q. Its absolute error is less than $1/Q$, and coefficients ± 1 stay exact. Lemma 6.3 measures the divergence exactly in $O(n \log(2n))$ queries. Thus the measured regularized gradient g obeys

$$g = \nabla \mathcal{F}(v) + \eta, \qquad \|\eta\|_2 \le \epsilon_0 := 2n^{3/2}/Q \le 1/(32\sqrt{n}). \tag{22}$$

The same canonical coefficient is used at both endpoints of a pair.

Initialize $x_0 = z_0 = 0$ and perform $J = 2T\lceil \log_2(16n) \rceil$ iterations of Nesterov's accelerated method for strongly convex functions [Nes83][Nes04, Section 2.2], with measured gradients:

$$\begin{aligned} v_j &= (x_j + \alpha z_j)/(1+\alpha), \\ g_j &= \text{measured approximate gradient at } v_j, \\ x_{j+1} &= v_j - g_j/L, \\ z_{j+1} &= (1-\alpha)z_j + \alpha v_j - \alpha g_j. \end{aligned} \tag{23}$$

All arithmetic and comparisons are rational. Since the measured divergence and β sum to zero, induction gives $\sum_v x_j(v) = \sum_v z_j(v) = \sum_v v_j(v) = 0$ exactly.

Lemma 7.3 (Inexact accelerated estimate)**.** *For $r = x_J$, the construction above satisfies*

$$\|\nabla \mathcal{F}(r)\|_2 \le \sqrt{2}\, T.$$

One final rounded-coefficient measurement gives $y = D(c_e h_e/Q)$ with

$$\|y - \beta + r\|_1 \le \sqrt{2n}\, T + 1/32. \tag{24}$$

The total query cost is $O(n\sqrt{nk}\log^2(2n))$.

Proof. Define $E_j = \mathcal{F}(x_j) - \mathcal{F}(r_*) + \|z_j - r_*\|_2^2/2$. The exact expansion, included in Section 16, gives

$$E_{j+1} \le (1-\alpha)E_j - \alpha\langle \eta_j, z_{j+1} - r_*\rangle.$$

Because $E_{j+1} \ge \|z_{j+1} - r_*\|_2^2/2$, it follows that

$$\sqrt{E_{j+1}} \le \sqrt{1-\alpha}\sqrt{E_j} + \alpha\epsilon_0\sqrt{2}.$$

By (21), $E_0 \le 5n^2$. Summing this contraction and using $1 - \sqrt{1-\alpha} \ge \alpha/2$ gives

$$\sqrt{E_J} \le e^{-J/(2T)}\sqrt{5}\, n + 2\sqrt{2}\epsilon_0 \le \frac{\sqrt{5}}{16} + \frac{\sqrt{2}}{16\sqrt{n}} < 1.$$

For an L-smooth convex function, $\|\nabla\mathcal{F}(r)\|_2^2 \le 2L(\mathcal{F}(r) - \mathcal{F}(r_*))$; this follows by comparing to one exact gradient step of length $1/L$. This proves the gradient bound. The final quantization error has ℓ_1 norm at most $2n^2/Q \le 1/32$, proving (24). There are $O(T\log(2n))$ calls, each costing $O(n\log(2n))$ queries. □

7.3 Learn the fractional pairs and round them

A pair with $|h_e| < Q$ was unsaturated before quantization, and hence belongs to $\mathcal{A}(r)$. Learn its capacity by querying its two-vertex shore; the known outdegrees and (17) determine c_e exactly. Lemma 7.1 bounds this cost by H. On such a pair the canonical capacity is $a_e = c_e(1 + h_e/Q)/2$ and the reverse capacity is $c_e - a_e$. Every other pair is already pure in a public direction.

Lemma 7.4 (Integral rounding on known pairs)**.** *The learned capacities can be rounded to integral $a'_e \in [0, c_e]$ so that, writing D_F for their incidence submatrix,*

$$\lfloor (D_F a)_v \rfloor \le (D_F a')_v \le \lceil (D_F a)_v \rceil \quad (v \in V).$$

The resulting integral representation has a known divergence b satisfying

$$\|b - y\|_1 \le 2n, \qquad e := b - \beta + r, \quad E := \|e\|_1 \le 8n\sqrt{k}. \tag{25}$$

Every positive represented arc $v \to w$ satisfies

$$r_w - r_v \le 2. \tag{26}$$

Finally, $q = (b - \beta)/2$ is an integral, zero-sum feasible transshipment in this representation, of total supply $R \le n^2$.

Proof. Add a root. When $(D_F a)_v > 0$, add a root-to-v arc carrying that value; when it is negative, add a v-to-root arc carrying its negative. Give each root arc integral lower and upper bounds equal to the floor and ceiling of its initial positive flow. Together with the fractional pair arcs, this is a feasible circulation with integer bounds. The undirected graph of nonintegral arcs cannot have a degree-one vertex, by conservation; if it is nonempty it has a cycle. Perturb along that cycle to the next integer boundary, preserving all bounds and previously integral arcs. At least one additional arc becomes integral. Repetition terminates and removing the root proves the coordinatewise rounding claim.

Only known pairs change. Their correction to signed divergence is $2D_F(a' - a)$, so its coordinates have absolute value at most two and are explicitly computable. Combine this with (24) to get

$$E \le 2n + \sqrt{2n}\,T + 1/32 \le 8n\sqrt{k},$$

using $T \le \sqrt{8nk} + 1$ and $n \ge 2$.

If an arc would increase potential by more than $1 + 1/k$, its clipped coefficient is saturated in the opposite direction, even allowing $|\omega_e| \le 1$. Quantization preserves that saturation, so before rounding no positive arc rises by more than $1 + 1/k \le 2$. On every pair rounded above, $|r_i - r_j - \omega_e| < 1/k$, so $|r_i - r_j| < 1 + 1/k \le 2$; either rounded direction satisfies (26).

Both b_v and β_v have the parity of the symmetrized degree, and both vectors sum to zero. Thus q is integral and zero-sum. On a canonical pair, compare the new capacity a'_e with the original capacity a^G_e. Send the signed amount $a'_e - a^G_e$. A positive amount fits the represented canonical capacity; a negative amount fits its represented reverse capacity. This flow has divergence $(b - \beta)/2$. Its total absolute mass is at most $\sum_e c_e \le n^2$, which also bounds total supply. The original directions are used only in this existence argument; the algorithm never reads them. □

8 Potential height bounds the repair volume

The next lemma is the geometric reason for the exponent 5/3. It permits fractional capacities and flows; integrality will enter only in the repair algorithm.

Lemma 8.1 (Potential height)**.** *Suppose a directed representation on n vertices has opposite pair capacities summing to at most two. Let r be a zero-sum real potential satisfying $r_w - r_v \le 2$ on every positive arc $v \to w$. If a transshipment with divergence $q = (-r + e)/2$ is feasible, write $E = \|e\|_1$ and $R = \sum_v q_v^+$. Then*

$$R \le 4n^{4/3} + \frac{5}{4}E. \tag{27}$$

Proof. Delete circulations from a feasible flow and decompose it into nonnegative source-to-demand paths. Sources with $r_v \ge 0$ supply at most $E/2$ in total, since there $q_v^+ \le e_v^+/2$. For $h \ge 4$, let

$H_h = \{v : r_v \geq h\}$. Its total demand is at least

$$\sum_{v \in H_h} q_v^- \geq \frac{1}{2} \sum_{v \in H_h} r_v - E/2.$$

Keep only paths starting at negative potential and ending in H_h. At most $E/2$ of the displayed demand was served by discarded sources, so the retained value Z_h satisfies

$$Z_h \geq \frac{1}{2} \sum_{v \in H_h} r_v - E. \tag{28}$$

If $Z_h > 0$, form an analysis network with all represented internal arcs, a supersource adjacent only to the retained negative-potential origins, and a supersink adjacent only from the retained high-potential endpoints. Give these auxiliary arcs the amounts prescribed by the retained paths. This flow of value Z_h is feasible. Every source-to-sink path, whether used by that decomposition or not, has internal potential gain greater than h and rises by at most two per internal arc. Its length d including the terminal arcs is therefore greater than $h/2 + 2$.

Consider BFS layers. Every outgoing arc of a layer prefix goes from its last layer to the next one. For middle prefixes, positive supersource arcs have already ended in layer one, and an earlier supersink arc would contradict distance d. Reverse terminal arcs cannot leave these prefixes. Thus only internal pairs cross them. Among at least $d/4$ disjoint middle layer pairs, one has total size at most $4n/d$. Its cut capacity is at most twice the product of the layer sizes, hence at most $8n^2/d^2$. Every feasible flow is bounded by this cut, so

$$Z_h \leq 8n^2/d^2 \leq 32n^2/h^2.$$

If $Z_h = 0$, equation (28) already bounds the tail by $2E$. In both cases,

$$\sum_{r_v \geq h} r_v \leq 2E + 64n^2/h^2, \qquad \sum_v r_v^+ \leq nh + 2E + 64n^2/h^2.$$

Since r and q have zero sum,

$$R = \frac{1}{4}\| - r + e\|_1 \leq \frac{nh}{2} + \frac{5}{4}E + \frac{32n^2}{h^2}.$$

Take $h = 4n^{1/3}$; the first and last terms each equal $2n^{4/3}$. The selected subflow and its analysis network need never be discovered. □

The lemma applies to the initial rounded representation, before repair. No claim is needed that (26) survives later residual reversals. With $k = \lceil n^{1/3} \rceil$, Lemmas 7.4 and 8.1 give

$$R \leq 4n^{4/3} + 10n\sqrt{k} = O(n^{4/3}). \tag{29}$$

This improves the volume bound using geometry of the feasible discrepancy, not an extra query or a hidden flow oracle.

9 Length-accounted blocking flows

We prove the repair theorem used above, including the cost of residual searches. We use Dinic's blocking flows [Din70] with the query accounting of Anand, Saranurak, and Wang [ASW25], who

compute a blocking flow f' with $\widetilde{O}(n + \mathrm{val}(f')\,\mathrm{dist}(s,t))$ queries by charging edge discoveries to augmenting-path length, and the layer bounds of Karzanov [Kar73] and Even and Tarjan [ET75]. Here they are applied to searches in an implicit representation. The representation may have any number of already known pairs, including split pairs. Every still-unknown pair is pure in a public direction and has capacity in $\{0,1,2\}$.

Theorem 9.1 (Integral discrepancy repair)**.** *For an explicit integral, zero-sum demand vector q with feasible total supply $1 \le R \le n^2$, a feasible flow can be discovered using*

$$O\Big(\Big(n\sqrt{R} + n\min\{R, n^{2/3}\}\Big)\log(2n)\Big) \subseteq O((n\sqrt{R} + n^{5/3})\log(2n))$$

outgoing-cut queries. For $R = 0$ no repair query is needed.

9.1 Legal residual searches

Add explicit arcs $s_0 \to v$ of capacity q_v^+ and $v \to t_0$ of capacity q_v^-. Their capacities may be large; their reverse residual arcs are also explicit. The maximum flow equals R by feasibility.

An undiscovered pair has never carried flow, so only its supplied public direction can have positive residual capacity. From a vertex v, first process all known residual arcs offline. Restrict the eligible endpoint set to unknown pairs whose public direction points out of v, and use (16). Bisection discovers a positive pair and its full capacity. Thereafter keep both residual capacities explicitly. This accesses the represented graph without knowing any original arc direction.

The next lemma is the blocking-phase bound of [ASW25], for searches in the represented graph.

Lemma 9.2 (One blocking phase)**.** *If the residual distance is d and an integral blocking phase sends value $a \ge 1$, it costs $O((n + da)\log(2n))$ queries.*

Proof. A complete residual BFS discovers at most $n + 1$ new vertices and has at most one final unsuccessful unknown-neighbor test per scanned vertex. It costs $O(n\log(2n))$ queries.

Run depth-first search on the frozen forward level graph. A vertex becomes permanently dead in the phase when no remaining route to the target exists from it. Eligible endpoints have the next level and are not dead. Crucially, return a complete source-to-target path before pushing any flow, then augment by its positive integral bottleneck. Repeat until the source is dead.

Charge each newly discovered pair by its first recursive traversal. A failed traversal makes its destination dead; distinct such failures have distinct destinations, so there are $O(n)$. A successful traversal returns a complete suffix which is propagated to the source and used in the very next augmentation. It is never an abandoned successful prefix. All augmentations use frozen forward levels. Newly created reverse arcs point backward and cannot cancel these phase contributions in the same phase. This statement concerns the phase augmentation, which may cancel flow from earlier phases.

Let g be the sum of the phase's augmentations and let $\ell(v)$ be the frozen level. Every used arc increases level by one; hence

$$\sum_{vw} g_{vw} = \sum_{vw} g_{vw}(\ell(w) - \ell(v)) = da. \tag{30}$$

This includes heavy terminal arcs. Every successful new pair carries at least one integral unit of the noncanceling phase flow, so there are at most da such pairs. Add $O(n)$ dead-destination discoveries and unsuccessful row tests. Every positive discovery costs $O(\log(2n))$ queries, proving the bound. Arbitrarily many scans of known arcs are free. □

The following is the layer bound of Karzanov [Kar73] and Even and Tarjan [ET75], in the form used by [ASW25].

Lemma 9.3 (Middle-layer deficit)**.** *If the remaining feasible flow deficit is D_0 and the finite residual distance is $d \geq 4$, then $D_0 \leq 8n^2/d^2$.*

Proof. For a prefix ending at layer i, an outgoing residual arc goes from L_i to L_{i+1}. If $1 \leq i \leq d-2$, no heavy terminal arc leaves this prefix: positive source arcs end at layer one, a positive sink arc would give a shorter target path, reverse source arcs point into layer zero, and reverse sink arcs start outside the prefix. Internal opposite residual capacities always sum to the original pair total, at most two. Therefore the prefix capacity is at most $2|L_i||L_{i+1}|$.

There are $\lfloor (d-1)/2 \rfloor \geq d/4$ disjoint pairs $(L_1, L_2), (L_3, L_4), \ldots$ of middle layers, containing at most n internal vertices. One has total size at most $4n/d$ and product at most $4n^2/d^2$. The corresponding cut bounds every remaining feasible flow. The same bound holds without auxiliary terminals, when all pairs are bounded. Known split pairs satisfy the same opposite-capacity invariant. □

Proof of Theorem 9.1. Dinic's [Din70] blocking phases strictly increase the residual distance: the old forward level graph has been blocked and new residual arcs point backward in its levels. Write the successful distances as $d_1 < \cdots < d_P$ and their integral augmentation values as $a_i \geq 1$. For an integer $j \geq 1$ let $H_j = \sum_{i:d_i \geq j} a_i$. This is the deficit before the first such phase, so $H_j \leq R$; for $j \geq 4$, Lemma 9.3 also gives $H_j \leq 8n^2/j^2$. The finite layer-cake identity yields

$$\sum_i d_i a_i = \sum_{j \geq 1} H_j \leq 7n\sqrt{R}.$$

Indeed, for $K = \lceil 4n/\sqrt{R} \rceil \geq 4$, the first K terms total at most $KR \leq 5n\sqrt{R}$, and the tail is at most $8n^2 \sum_{j>K} j^{-2} \leq 8n^2/K \leq 2n\sqrt{R}$.

As in the classical phase bound [Kar73, ET75], put $h = \max\{4, \lceil n^{2/3} \rceil\}$. After h successful phases any next distance exceeds h, leaving at most $8n^2/h^2$ integral units. Each subsequent phase sends at least one unit, so $P = O(n^{2/3})$ and also $P \leq R$. Sum Lemma 9.2 to obtain the theorem. Feasible repair stops at the known value R; it needs no query to an auxiliary-network cut. □

9.2 Cut equivalence and the terminal solve

After repair, replace each discovered pair by its residual capacities. If its flow divergence was q, the represented divergence becomes $b - 2q = \beta$; the pair totals remain c_e. Lemma 6.1 proves cut equivalence. Every unknown pair is still pure in a public direction, and every known pair has explicit residual capacities.

If desired, the split pairs can be compressed to a forest offline. An integral split pair has total capacity two and split $1 + 1$. On an undirected cycle of split pairs, add one unit around a cyclic orientation and subtract it in the opposite directions. This preserves divergence and pair totals and makes all pairs of that cycle pure. Repeat until the split support is acyclic. This optional operation is performed after the height argument, and is not needed for the residual search bound.

Run the same blocking-flow algorithm between s, t in the resulting representation, stopping when BFS cannot reach t. Its actual maximum value λ need not be known. The preceding proof applies with total augmentation λ, and one final unsuccessful BFS costs $O(n \log(2n))$. Cut equivalence now justifies $\lambda \leq F(\{s\}) \leq n - 1$. Thus the terminal solve costs

$$O((n\sqrt{\lambda} + n^{5/3} + n) \log(2n)) = O(n^{5/3} \log(2n)).$$

By the max-flow min-cut theorem [FF56], the final reachable shore is a minimizing shore in the representation and therefore in G. One original query returns its exact value.

Proof of Theorem 1.1. Choose $k = \lceil n^{1/3} \rceil$. Lemmas 7.1–7.4 construct the public rounded representation in

$$O\left(n\sqrt{nk}\log^2(2n) + \frac{n^2}{k} + n\log(2nk)\right) = O(n^{5/3}\log^2(2n))$$

queries. Equation (29) gives $R = O(n^{4/3})$, since $n\sqrt{k} = O(n^{7/6})$. The repair and the terminal solve each cost $O(n^{5/3}\log(2n))$ additional queries. All preprocessing was public and deterministic, and the final shore and value are exact. □

10 The randomized engine

The graph primitive concerns weights assigned by a public formula, rather than unknown input weights. This distinction allows the changing normal matrices of an LP to be accessed through the same unweighted cut oracle.

Theorem 10.1 (Publicly weighted spectral primitive)**.** *Suppose $c_e \in \{0, \dots, C\}$ and a public formula assigns a nonnegative weight w_e to every potential pair. Positive weights can be evaluated, compared, and output exactly. For $0 < \theta, \zeta \le 1/2$, an explicit weighted graph Q satisfying*

$$(1-\theta)L \preceq L_Q \preceq (1+\theta)L, \qquad L = \sum_e c_e w_e d_e d_e^\top,$$

is obtained with probability at least $1-\zeta$ using at most

$$O\left(C\theta^{-2} n \log^8 \frac{2Cn}{\theta\zeta}\right)$$

queries on every execution. The guarantee holds conditional on any earlier transcript if the call uses fresh coins. The query bound is independent of the ratio of positive public weights.

Proof. Section 14 gives the construction, including its query cap. It uses public filtered searches to construct disjoint weighted spanners, then recursive pair sampling with a matrix Bernstein bound. □

Adding a known positive semidefinite matrix E preserves the same relative approximation to $E + L$. Hidden incidence rows with common per-pair coefficient ω_e contribute $\sum_e c_e \omega_e a_e a_e^\top$, where a_e is the incidence column. Thus explicit rows may be processed offline while the hidden Gram matrix is handled spectrally.

Theorem 10.2 (Repaired implicit incidence LP)**.** *Consider a finite-box LP $\min c^\top x$ subject to $A^\top x = b$ and $\ell \le x \le u$, with $A \in \mathbb{R}^{M \times n}$. Its hidden rows are incidence vectors of the unknown symmetrization, with constant pair multiplicity, identical public data on duplicate copies, and $O(n)$ explicit rows including $+I_n$. Suppose all rows are nonzero with norm at most $\sqrt{2}$, the dyadic input data have polynomial numerical bounds, a strictly feasible point of inverse-polynomial minimum slack is given by public formulas, and a polynomial lower bound on row norms is known. For inverse-polynomial $0 < \varepsilon \le 1$, the LP has a randomized algorithm returning an exactly feasible ε-suboptimal point, with public formulas for its hidden coordinates, using $\widetilde{O}(n^{3/2})$ queries with high probability.*

Proof. The full replacement execution is given in Section 15. Its normal systems are hidden publicly weighted Laplacians plus explicit positive semidefinite matrices. Theorem 10.1 gives preconditioners, and Lemma 6.3 gives exact products and feasibility repairs. The corrected regularizer is $1/(2\overline{M})$; that section proves the sensitivity and projection inequalities and propagates the changed slack and normal-system bounds throughout the execution. There are $\widetilde{O}(\sqrt{n})$ batches of $\widetilde{O}(n)$ queries. □

Corollary 10.3 (Randomized terminal bound)**.** *A minimum terminal cut of a simple unweighted digraph can be found with high probability in $\widetilde{O}(n^{3/2})$ outgoing-cut queries.*

The next two sections prove the reduction to Theorem 10.2. They are valid for any algorithm meeting that explicitly stated interface.

Theorem 10.4 (Global reduction)**.** *Suppose terminal mincut in the same model costs $Q_{st}(n)$ queries with high probability. For every integer $1 \le h \le n-1$, global mincut costs*

$$\widetilde{O}\left(nh + \frac{n}{h} Q_{st}(n)\right)$$

queries with high probability. Every terminal instance uses the original graph. In particular $h = \min\{n-1, \lceil n^{3/4} \rceil\}$ and Corollary 10.3 give $\widetilde{O}(n^{7/4})$ queries.

11 A penalized incidence LP for terminal cuts

Fix $s \neq t$, put $\chi = e_s - e_t$, and write $\lambda = \lambda_{st}(G)$. The hidden variables are $z_{uv,j}$, one for each of the c_{uv} copies. Let $a_{uv} = e_u - e_v$ for $u < v$, and let B have one row $a_{uv}^\top$ per copy. Add a variable q and two explicit nonnegative variables y_v^+, y_v^- per vertex. The hidden copies, the star variables, and their penalty in the objective come from the reachability LP of [Anon26, eq. (4)]; the flow variable q is added here. Consider

$$\begin{aligned} \text{minimize} \quad & R - q/2, && R = \sum_v (y_v^+ + y_v^-), \\ \text{subject to} \quad & B^\top z + q\chi + y^+ - y^- = \beta, \\ & -1 \le z_{uv,j} \le 1, \quad -2 \le q \le 2n, \\ & 0 \le y_v^+, y_v^- \le 4n. \end{aligned} \tag{31}$$

Write $f = q/2$ and $r = y^+ - y^-$. Each hidden copy carries fractional arc capacities $(1 + z_{uv,j})/2$ from u to v and $(1 - z_{uv,j})/2$ in the reverse direction. Call this fractional digraph P_z. Its capacities are nonnegative, its symmetrization is U, and its divergence is $\beta - 2f\chi - r$. Therefore

$$P_z(S) = F(S) - f\chi(S) - r(S)/2 \qquad \text{for every } S \subseteq V. \tag{32}$$

Lemma 11.1. *The optimum of (31) is $-\lambda$. Every feasible point with objective at most $-\lambda + \varepsilon$ satisfies*

$$R \le 2\varepsilon, \qquad |f - \lambda| \le \varepsilon.$$

Proof. Let S_* be a minimum s–t cut. Nonnegativity of $P_z(S_*)$ and (32) give

$$f \le \lambda - r(S_*)/2 \le \lambda + R/2.$$

Consequently $R - f \ge -\lambda + R/2 \ge -\lambda$ for every feasible point, including points with negative f.

Conversely, take any actual maximum flow of value λ in G, and form its residual network. On each unordered pair, the sum of the two residual capacities is exactly c_{uv}, and its divergence is $\beta - 2\lambda\chi$. Represent these capacities by a common z_{uv} on all copies of the pair. This gives a feasible point of (31) with $q = 2\lambda$ and $y^+ = y^- = 0$. The bound $\lambda \le n - 1$ ensures that the q box is respected. Its objective is $-\lambda$. This is an existence argument; the algorithm does not need the flow.

For a near-optimal point, $R - f \le -\lambda + \varepsilon$, so the preceding lower bound implies $R \le 2\varepsilon$. Also $f \ge \lambda + R - \varepsilon \ge \lambda - \varepsilon$, while $f \le \lambda + R/2 \le \lambda + \varepsilon$. □

Checking every LP input requirement. The complete row matrix is $A = [B; \chi^\top; I_n; -I_n]$ and has rank n. There are $2n+1$ explicit rows, all with norm at most $\sqrt{2}$; the hidden pair multiplicity is at most two. Hidden bounds are $[-1, 1]$, their objective coefficient is zero, and their initial coordinate is zero. Copies of a pair have identical data. A known strict feasible start is

$$z^0 = 0, \quad q^0 = 0, \quad (y_v^+)^0 = 2n + \beta_v/2, \quad (y_v^-)^0 = 2n - \beta_v/2.$$

All these data are dyadic with $O(\log n)$ bits and polynomial magnitude. Initial slacks and box widths have polynomial bounds. The number of rows is $M = |E| + 2n + 1 = O(n^2)$ and is known from $\sum_v d^+(v)$. The accuracy used below is inverse polynomial. Thus this instance meets Theorem 10.2, including the explicitly supplied feasible start. The implicit solver yields a common publicly evaluable z_{uv} for each potential pair, without querying whether that pair is present.

12 Extracting the exact cut without another LP

Choose dyadic parameters

$$\tau = 2^{-\lceil \log_2(8n^2) \rceil}, \qquad \varepsilon = \tau/16.$$

Apply Theorem 10.2 to (31) with additive error ε. In the following argument condition on its success event. For an ordered pair u, v, let p_{uv} be the per-copy fractional arc capacity obtained from the common $z_{\min(u,v),\max(u,v)}$. Evaluate a public dyadic $\widehat{p}_{uv}$ with error at most $\tau/4$. Apply clipping uniformly to potential-pair formulas if needed to make them total; the clipping leaves valid actual-pair capacities unchanged. As in the threshold digraph of [Anon26, eq. (8)], define an implicit directed graph J by

$$(u, v) \in E(J) \quad \Longleftrightarrow \quad c_{uv} > 0 \ \text{ and } \ \widehat{p}_{uv} > \tau.$$

Lemma 12.1. *The set W of vertices reachable from s in J excludes t and is an exact minimum s–t cut in G. It can be found using $O(n \log n)$ outgoing-cut queries.*

Proof. For the true minimum cut S_*, Lemma 11.1 gives

$$0 \le P_z(S_*) = \lambda - f - r(S_*)/2 \le 2\varepsilon = \tau/8.$$

Every per-copy arc leaving S_* is at most this total, so its rounded value is at most $\tau/8 + \tau/4 < \tau$. No arc of J leaves S_*. Therefore $s \in W \subseteq S_*$ and $t \notin W$.

No arc of J leaves W. Hence every actual per-copy arc leaving W has $p_{uv} \le 5\tau/4$. Since there are at most $n(n-1)$ copies,

$$P_z(W) \le \frac{5}{4} n(n-1)\tau < \frac{5}{32}.$$

Using (32) again,

$$F(W) = P_z(W) + f + r(W)/2 < \lambda + 2\varepsilon + \frac{5}{32} < \lambda + 1.$$

Integrality and the fact that W separates s, t imply $F(W) = \lambda$.

The implementation is the implicit search of [Anon26, Lemma 3.4]. During a graph search let T be the currently undiscovered vertices satisfying $\widehat{p}_{uv} > \tau$. This is a publicly computable set. Equation (16) tests whether there is an actual neighbor of u in T, and binary search finds one. Each successful search discovers a new vertex. There are at most $n-1$ successes and n final failed searches, giving $O(n \log n)$ queries. One final query evaluates $F(W)$ exactly. □

This proves Corollary 10.3. On an unsuccessful LP execution we may return a trivial terminal shore. If the search reaches t, return $\{s\}$ instead. Every candidate is thus an actual terminal cut and its value is checked with the oracle. Independent repetitions and taking the least checked value amplify success without allowing an erroneous answer below the true optimum.

13 Small shores and the global reduction

The reduction balances a local step for small shores against terminal calls on sampled vertices, as in the directed minimum-cut algorithm of Cen, Li, Nanongkai, Panigrahi, and Saranurak [CLN+21]. In the cut-query model the local step uses learned pair totals instead of original arcs.

Set

$$\delta = \min_{v \in V} \{d^+(v), d^-(v)\}.$$

A known singleton or co-singleton cut attains δ. If $\delta = 0$ we are done. Fix $1 \le h \le n-1$.

Lemma 13.1 (Learning enough for all small shores)**.** *Using $O(nh \log n)$ queries, one can produce a cut of value $L \le \delta$ such that $L \le F(S)$ whenever S is nonempty and proper and $\min\{|S|, |V \setminus S|\} \le h$.*

Proof. If $\delta > h$, no learning is needed: take the cut of value δ. Indeed, if $s = |S| \le h$, simplicity gives

$$F(S) \ge s\delta - s(s-1) = \delta + (s-1)(\delta - s) \ge \delta.$$

The same calculation using indegrees applies if $|V \setminus S| \le h$.

Now suppose $\delta \le h$, and put

$$D_+ = \{v : d^+(v) \le \delta + h\}, \qquad D_- = \{v : d^-(v) \le \delta + h\}.$$

Learn all multiplicities of $U[D_+]$ and $U[D_-]$ using the row query (16) and binary search. The number of copies inside D_+ is at most $\sum_{v \in D_+} d^+(v) \le n(\delta + h) \le 2nh$. For D_- use the identical argument with indegrees. The total query cost is $O(nh \log n)$; no original arc directions are learned.

Equation (17) now evaluates $F(S)$ without queries for every $S \subseteq D_+$. Likewise

$$F(V \setminus T) = \sum_{v \in T} d^-(v) - \sum_{\{u,v\} \subseteq T} c_{uv}$$

evaluates every $T \subseteq D_-$. Enumerate these subsets offline, excluding empty and full shores, and keep their least cut together with the cut of value δ.

To see coverage, consider any cut with $|S| \le h$ and $F(S) < \delta$. Every $v \in S$ has $d^+(v) \le |S| - 1 + F(S) \le h - 1 + \delta$, so $S \subseteq D_+$. If instead $|V \setminus S| \le h$, the same argument with indegrees puts the complement inside D_-. Cuts of value at least δ are already covered by the initial candidate. □

Proof of Theorem 10.4. Run Lemma 13.1. Fix a root a, and independently sample $k = \lceil c(n/h) \log n \rceil$ uniform vertices, with replacement. For every sampled $v \ne a$, run the terminal routine on (a, v) and (v, a). Return the least of all checked cut values and the local candidate.

Fix one global minimum cut S_*. If it has a shore of size at most h, the local candidate is already optimal. Otherwise both shores have size greater than h. Each sample lands on the opposite side from a with probability greater than h/n. Thus the probability of missing that side in all samples is at most $(1-h/n)^k \le n^{-c}$. For a successful sample one of the two ordered terminal pairs is separated in the direction of S_*. Its terminal mincut equals the global optimum. Amplifying the terminal routines to error n^{-c-3} and taking a union bound completes the success argument. The query cost is $O(nh \log n) + \widetilde{O}((n/h)Q_{st}(n))$. □

14 Weighted spectral sparsification with public pair weights

This section supplies the full graph-oracle construction underlying Theorem 10.1. Matrix Bernstein is the only external concentration theorem used.

14.1 Statement and oracle model

Let $V = [n]$, $n \ge 2$, and let G be a hidden loopless directed multigraph. Its oracle returns the number of outgoing arcs, with multiplicity,

$$F(S) = |E_G(S, V \setminus S)|.$$

For an unordered pair $e = \{u, v\}$, let c_e count all arcs between its endpoints, in both directions. Assume $c_e \in \{0, 1, \dots, C\}$ for known $C \ge 1$. For simple digraphs allowing antiparallel arcs, $C = 2$. Orient each pair canonically and put $a_e = e_u - e_v$. A public formula assigns a nonnegative weight w_e to every possible pair, including nonedges. Positive weights can be evaluated, compared exactly, and output. In the LP application they are positive dyadic numbers with polynomial bit bounds.

The input may depend on previous oracle answers, but its pair-weight formula is fixed throughout this call. A *public predicate* on pairs is computable from labels, that formula, recorded answers, and public randomness, without a new adjacency lookup. Only calls to F are charged; computation, storage, and random bits are unrestricted.

Theorem 14.1 (Spectral primitive)**.** *For $0 < \varepsilon, \zeta \le 1/2$, there is a randomized algorithm using at most*

$$O\left(C\varepsilon^{-2} n \log^8 \frac{2Cn}{\varepsilon\zeta}\right)$$

outgoing-cut queries on every execution. With probability at least $1-\zeta$, it outputs an explicit weighted graph Q satisfying

$$(1-\varepsilon)L \preceq L_Q \preceq (1+\varepsilon)L, \qquad L = \sum_e c_e w_e a_e a_e^\top. \tag{33}$$

Every nonfailure output has at most $O(C\varepsilon^{-2} n \log^6(2Cn/(\varepsilon\zeta)))$ pairs, all of positive input conductance. Neither bound depends on $\max_{w_e>0} w_e / \min_{w_e>0} w_e$. The guarantee holds conditional on any earlier transcript when this call uses fresh randomness.

The construction specializes weighted-spanner sampling [BS07, Kou14, KP12] to this oracle; the spanner bundles bound effective resistances, which control the sampling probabilities as in Spielman and Srivastava [SS11]. Section 4 of the external note [Anon26] proposes this approach; we prove its needed ingredients here, using fair cluster coins and explicit public pair masks. There is no appeal to the disputed implicit-LP stability claim. The algorithm may output FAIL at a stated cap. Spectral failures need not be recognizable, so the theorem is a Monte Carlo guarantee.

14.2 The queries needed to see a minimum edge

For disjoint $A, B \subseteq V$, cancellation of arcs gives

$$\sum_{u \in A,\, v \in B} c_{uv} = F(A) + F(B) - F(A \cup B). \tag{34}$$

In particular, for $v \notin Y$, three queries give $N(v, Y) := \sum_{u \in Y} c_{uv}$. This count is zero precisely when no actual pair joins v to Y.

Lemma 14.2 (Filtered minimum search)**.** *Fix a public support predicate P, a vertex v, and any public ordered list of eligible endpoints. The minimum eligible actual pair vu satisfying $P(vu)$, or the fact that none exists, can be found in $O(\log n)$ cut queries. Its multiplicity can also be learned within this bound.*

Proof. Form the list of eligible endpoints satisfying P, which requires no adjacency information. One count tests the entire list. If it is nonempty in the graph, test the left half and retain it if its count is positive; otherwise retain the right half. The retained interval always contains the first actual endpoint in the original order. After at most $\lceil \log_2 n \rceil$ tests, a singleton remains. Its count is its multiplicity. Empty lists need no query. □

Give each pair with $w_e > 0$ length $\ell_e = 1/w_e$, and order pairs by $\mathrm{key}(e) = (\ell_e, \mathrm{ID}(e))$, with a fixed unique ID. Lemma 14.2 finds minimum-key edges in any publicly specified cluster or union of clusters. Even nonedges may occur anywhere in the sorted candidate list. Sorting all potential pairs is free in this query model. No search over numerical weight scales is used.

14.3 A weighted spanner with a query cap

Fix a public predicate P supported on positive-weight pairs. Write $E^P = \{e : c_e > 0,\ P(e) = 1\}$. We construct a spanner for lengths ℓ_e, ignoring multiplicity for this step. Set

$$k = 1 + \lceil \log_2 n \rceil, \qquad \alpha = 2k - 1.$$

A trial has k phases. At the beginning of phase i, let E_i denote the actual residual pairs, $\mathcal{C}_i$ the clusters of active vertices, and H_i the pairs already selected. The residual is stored only as a public predicate. Initially $E_1 = E^P$, the clusters are singletons, and $H_1 = \varnothing$. Every cluster has a center which is one of its members.

At all times residual pairs join distinct active clusters. An inactive vertex has no actual residual incident pair. A phase is as follows.

1. Freeze the old residual predicate and all old cluster labels. For $i < k$, sample every old cluster independently with probability $1/2$; write $\mathcal{S}_i$ for the sampled family. In phase k, take $\mathcal{S}_k = \varnothing$.
2. A vertex in a sampled old cluster keeps its label and makes no reports. For each other active vertex v, let f_v be its minimum-key residual edge to a sampled old cluster, if there is one. Put $\sigma_v = \mathrm{key}(f_v)$, or $+\infty$ if there is none. For each old cluster D having an edge to v, let $f(v, D)$ be its minimum-key such edge. Vertex v reports precisely those clusters with $\mathrm{key}(f(v, D)) \le \sigma_v$.
3. For each report (v, D), select $f(v, D)$ and record the *entire old block* $E_i(v, D) = \{vu \in E_i : u \in D\}$ for deletion. For each processed, unsampled vertex v, if f_v exists, give v the new label and center of its sampled old cluster. Otherwise v becomes inactive.
4. Only after all reports are computed, apply their union of block deletions and the new labels. Delete any pair whose endpoints have the same new label. Proceed to the next phase.

The final phase has no sampled cluster, so every active vertex reports every neighboring cluster and no actual residual pair remains. Each selected pair is retained in the spanner, irrespective of its subsequent removal from the residual.

Oracle implementation. Find f_v by Lemma 14.2. To enumerate the reports, repeatedly find the minimum eligible residual edge with key at most σ_v into an old cluster not yet reported by this vertex. Its endpoint identifies the next reported old cluster; exclude that entire cluster from this vertex's next search. If $\sigma_v = +\infty$, omit the key restriction. The first edge encountered in a cluster is its minimum edge. Thus exactly the specified reports are generated, using one initial search, one per report, and one terminating search per vertex.

These exclusions are local to the reporting vertex. The global residual predicate is unchanged until the phase ends. For a possible pair uv, the block-deletion predicate then asks whether the old cluster of u was reported by v, or vice versa. This and the new-label test are public predicates, including on nonedges. Hence every query is licensed by (34); the procedure never needs to enumerate all actual pairs in a reported block.

Lemma 14.3 (Stretch for every completed trial)**.** *For every sequence of cluster choices, the completed trial is an α-spanner of E^P.*

Proof. We prove the following stronger invariant. For every $e = uv \in E_i$ and each endpoint $x \in \{u, v\}$, there is an H_i-path from x to its old center with at most $i - 1$ edges, each of length at most ℓ_e. It holds initially with empty paths.

First consider an edge $e = vu$ surviving all phase-i block deletions, before the deletion of edges with equal new labels. Vertex v cannot become inactive: in that case it would have reported all its neighboring old clusters and deleted e. If its old cluster is sampled, its center path is unchanged. Otherwise it joins a sampled cluster by $f_v = vy$. Since it did not report the old cluster D of u,

$$\mathrm{key}(f_v) < \mathrm{key}(f(v, D)) \le \mathrm{key}(e).$$

Thus $\ell_{f_v} \le \ell_e$. The old invariant applied to f_v supplies a y-to-center path of at most $i - 1$ edges, each of length at most ℓ_{f_v}. Prepending the selected edge f_v and removing cycles if necessary gives the new center path for v, with at most i edges of length at most ℓ_e. This proves the next invariant after same-label deletions.

Next consider an edge $e = vu$ deleted through a reported block $E_i(v, D)$. Put $f = f(v, D) = vy$. Both u and y have old-center paths to the center of D, of at most $i - 1$ edges. For u, apply the old invariant to e; for y, apply it to f, and use $\ell_f \le \ell_e$. Together with f, these paths give a v-to-u walk of length at most $(2i - 1)\ell_e$. Removing cycles gives a path.

If e survives the block deletions but receives equal new labels, its two new center paths instead give length at most $2i\ell_e$. Such an edge can occur only in a nonfinal phase, so $2i \le 2k - 2$. In the final phase all remaining pairs are removed through reported blocks. Every original pair therefore has a replacement of length at most $(2k - 1)\ell_e$, and all its selected edges remain in the output. □

Lemma 14.4 (Size, queries, and amplification)**.** *For any $0 < \beta < 1$, an α-spanner of E^P with at most $4kn$ pairs can be found, or* FAIL *returned, using*

$$O(n \log^2 n\,(1 + \log(1/\beta)))$$

queries on every execution. The failure probability is at most β, conditional on all prior information.

Proof. Analyze an uncapped trial. Condition on the start of a nonfinal phase. Order the neighboring old clusters of a fixed vertex v by their minimum incident key. Its own old cluster is absent from this list, because the residual has no intracluster pairs. If its own cluster is sampled, it reports nothing. Otherwise, its report count is the position of the first sampled neighboring cluster, truncated at the number of neighbors. The neighboring-cluster coins are independent fair coins even after conditioning on the own-cluster coin. Hence

$$\Pr\{\text{reports by } v \geq j \mid \text{old state}\} \leq 2^{-(j-1)}, \qquad \mathbb{E}[\text{reports by } v \mid \text{old state}] \leq 2.$$

Each sampled old cluster persists, since all of its old members stay; each unsampled old cluster disappears. Consequently

$$\mathbb{E}|\mathcal{C}_k| = n2^{-(k-1)} \leq 1.$$

There are at most $n|\mathcal{C}_k|$ reports in the final phase. If N_{rep} is the total report count, then

$$\mathbb{E}N_{\text{rep}} \leq 2n(k-1) + n \leq 2kn.$$

Stop a trial when it exceeds $4kn$ reports. Markov's inequality gives failure probability at most $1/2$; a completed trial has the stretch proved above. The capped trial has $O(kn + N_{\text{rep}}) = O(kn)$ searches, each costing $O(\log n)$ queries. Learning each selected pair's multiplicity adds at most a constant number of queries per report.

Use $J = \lceil \log_2(1/\beta) \rceil$ fresh capped trials, returning the first completed one. Each restart uses the original predicate P, and discards only the failed trial's tentative spanner. Past query answers may be retained, but impose no restriction on the fresh cluster coins. The same uniform conditional failure bound therefore gives probability at most $2^{-J} \leq \beta$ that every trial fails. The query cap holds even on those executions. □

14.4 Disjoint spanners certify small pair leverage

Fix any actual support E' represented by a public predicate, and any common scale $s > 0$. Its conductances are $\omega_e = sc_e w_e$, and its Laplacian is L'. Construct t successive α-spanners $S_1, \ldots, S_t$, deleting all pairs selected in earlier spanners before the next call. Let $B = \bigcup_j S_j$. The deletions are public because every selected pair is explicit. Only selected pairs are deleted between spanner calls. The block deletions inside a trial are its temporary search state and are discarded when that call ends. If the actual residual becomes empty, subsequent spanners may be empty.

Lemma 14.5 (Grouped leverage bound)**.** *If $e \in E' \setminus B$, then*

$$\omega_e a_e^\top (L')^\dagger a_e \leq \frac{C\alpha}{t}. \tag{35}$$

Proof. The pair $e = uv$ was present at the start of every spanner call. For each j, choose a simple u-to-v path $P_j \subseteq S_j$ with $\sum_{f \in P_j} 1/w_f \leq \alpha/w_e$. These paths share no unordered pair, because the spanners do not. They may share vertices.

Send $1/t$ units of flow along each path, giving a unit u-to-v flow q. The resistance of pair f is $1/(sc_f w_f) \leq 1/(sw_f)$. Pair disjointness therefore gives

$$\sum_f \frac{q_f^2}{sc_f w_f} \leq \frac{1}{t^2 s} \sum_{j=1}^{t} \sum_{f \in P_j} \frac{1}{w_f} \leq \frac{\alpha}{tsw_e}. \tag{36}$$

For completeness, the minimum energy of a unit u-to-v flow is $a_e^\top (L')^\dagger a_e$. If D is a matrix of oriented incidence columns and $W = \operatorname{diag}(\omega)$, then $L' = DWD^\top$. The flow $q_* = WD^\top (L')^\dagger a_e$ satisfies

$Dq_* = a_e$, since $a_e \in \mathrm{range}(L')$. For any other unit flow, write $q = q_* + z$, $Dz = 0$. Its cross term is $q_*^\top W^{-1} z = a_e^\top (L')^\dagger Dz = 0$, so its energy is the energy of q_* plus a nonnegative term. The energy of q_* is exactly $a_e^\top (L')^\dagger a_e$. Apply this identity to (36), then multiply by $sc_e w_e$ and use $c_e \le C$. □

The entire multiplicity of a pair will be sampled as one block. That is why its leverage involves c_e, and why the upper bound has a factor C. There is no need to distinguish hidden copies.

14.5 One round of sampling

Keep B with its current conductances. For each remaining pair, independently choose $\xi_e \in \{0, 1\}$ with $\Pr(\xi_e = 1) = 1/4$. If it survives, multiply its conductance by four. Denote the resulting Laplacian by $\widehat{L}$.

Lemma 14.6 (One-round spectral error)**.** *Conditional on a completed bundle and all earlier information, for* $0 < \delta \le 1$,

$$\Pr\{(1-\delta)L' \not\preceq \widehat{L} \text{ or } \widehat{L} \not\preceq (1+\delta)L'\} \le 2n \exp\left(-\frac{\delta^2 t}{8C\alpha}\right). \tag{37}$$

Proof. If $L' = 0$, the claim is immediate. Otherwise restrict to $\mathrm{range}(L')$, and write

$$A_e = (L')^{\dagger/2}(\omega_e a_e a_e^\top)(L')^{\dagger/2}, \qquad \lambda = \frac{C\alpha}{t}.$$

For every pair outside B, A_e is positive semidefinite of rank one, and Lemma 14.5 gives $\|A_e\| \le \lambda$. Also $\sum_{e \notin B} A_e \preceq I$, because the sum over *all* pairs of the current graph is the identity on its range. The normalized error is a sum of independent centered matrices

$$(L')^{\dagger/2}(\widehat{L} - L')(L')^{\dagger/2} = \sum_{e \notin B} Z_e, \qquad Z_e = (4\xi_e - 1)A_e.$$

Since $\mathbb{E}(4\xi_e - 1)^2 = 3$ and $A_e^2 \preceq \lambda A_e$,

$$\|Z_e\| \le 3\lambda, \qquad \left\| \sum_e \mathbb{E} Z_e^2 \right\| \le \left\| 3\lambda \sum_{e \notin B} A_e \right\| \le 3\lambda.$$

The two-sided matrix Bernstein inequality [Tro12, Theorem 6.1(ii)] states that independent centered self-adjoint d-dimensional matrices with norm at most b and total variance norm at most v satisfy

$$\Pr\left\{ \left\| \sum_e Z_e \right\| > x \right\} \le 2d \exp\left(-\frac{x^2}{2(v + bx/3)}\right).$$

With $d \le n$, $b = v = 3\lambda$, and $x = \delta \le 1$, the denominator is at most 8λ, proving the probability bound. Finally, every retained pair belonged to the current graph, so both L' and $\widehat{L}$ vanish on $\ker L'$. The normalized norm bound on the range is therefore equivalent to the two Loewner inequalities on all of $\mathbb{R}^n$. □

Notice that normalization uses the full current graph, including B. We never require the sampled complement to be connected or invert its Laplacian. Nor does the algorithm compute any resistance: resistances are used only in the proof.

14.6 Recursion, failure probability, and query accounting

Choose the following public parameters:

$$R = \left\lceil \log_4 \frac{4n^2}{\zeta} \right\rceil, \qquad \delta = \frac{\varepsilon}{3R}, \tag{38}$$
$$t = \left\lceil 8C\alpha\delta^{-2} \log \frac{8nR}{\zeta} \right\rceil, \qquad \beta = \frac{\zeta}{4Rt}.$$

Use natural logarithms in the expression for t. At recursion level $r \in \{0, \dots, R\}$, the support has a public predicate P_r and the actual current Laplacian is

$$L_r = \sum_{e:\, P_r(e)=1} c_e 4^r w_e a_e a_e^\top.$$

Initially $P_0(e)$ means $w_e > 0$. At each $r < R$:

1. Construct t successive spanners by Lemma 14.4, each with failure probability at most β, deleting each completed spanner before the next. Let B_r be their union. If any call returns FAIL, return FAIL for the whole procedure.
2. Store each pair $e \in B_r$ in the output with weight $c_e 4^r w_e$, learning c_e by a singleton count if it has not already been learned. These stored weights are never rescaled later.
3. After the bundle is fixed, draw a fresh independent fair pair of bits for every possible unordered pair. Let $h_r(e) = 1$ exactly when both are one, and define

$$P_{r+1}(e) = P_r(e)\,[e \notin B_r]\,[h_r(e) = 1]. \tag{39}$$

   Continue on this predicate at scale 4^{r+1}.

At depth R, test whether the actual residual is empty, with one filtered-neighbor count per vertex. If not, return FAIL. Otherwise return the explicit union of the stored bundles.

The masks in (39) can be stored explicitly for all potential pairs. Their generation costs no cut queries. This uses full independence, with no assertion that a short hash seed provides it. The spanner's base lengths remain $1/w_e$ at every level, since the common conductance scale does not change their stretch.

Proof of Theorem 14.1. There are at most Rt capped spanner calls, so their failures have total probability at most $Rt\beta = \zeta/4$, by conditioning on the transcript before each call. On every history that reaches a completed bundle, its pair mask is still fresh. Lemma 14.6 and the choice of t give conditional one-round failure probability at most $\zeta/(4R)$. Thus the probability of any failed round is at most $\zeta/4$.

For the terminal check, fix an original pair. Given the entire history through the current bundle, its probability of surviving the next level is either zero, if deleted, or $1/4$, if retained for sampling. By induction its probability of surviving R levels is at most 4^{-R}. This remains true with adaptive bundles; if the algorithm aborts, define subsequent residuals to be empty. The union bound over fewer than n^2 pairs gives

$$\Pr\{\text{nonempty terminal residual on an unaborted run}\} \le n^2 4^{-R} \le \zeta/4.$$

Thus all needed events hold with probability at least $1 - 3\zeta/4 \ge 1 - \zeta$.

Let K_r be the Laplacian of bundle B_r, stored at its level's scale. On a successful one-round event,

$$(1-\delta)L_r \preceq K_r + L_{r+1} \preceq (1+\delta)L_r. \tag{40}$$

The terminal residual is zero. Prove the output approximation backwards. If the later output Q_{r+1} satisfies $aL_{r+1} \preceq L_{Q_{r+1}} \preceq bL_{r+1}$, where $a \le 1 \le b$, positivity of K_r gives

$$a(K_r + L_{r+1}) \preceq K_r + L_{Q_{r+1}} \preceq b(K_r + L_{r+1}).$$

Combine with (40). Starting with the exact zero residual, the factors at the root are $(1-\delta)^R$ and $(1+\delta)^R$. Bernoulli's inequality and $e^x \le 1+2x$ for $0 \le x \le 1/6$ show

$$(1-\delta)^R \ge 1 - R\delta \ge 1-\varepsilon, \qquad (1+\delta)^R \le e^{\varepsilon/3} \le 1 + 2\varepsilon/3 \le 1+\varepsilon.$$

This proves (33), including disconnected inputs.

For the cost, put $D = \log(2Cn/(\varepsilon\zeta))$. Then $k, \alpha, R = O(D)$, and

$$t = O(C\varepsilon^{-2}D^4), \qquad Rt = O(C\varepsilon^{-2}D^5), \qquad \log(1/\beta) = O(D).$$

Each spanner call costs $O(nD^3)$ cut queries on every execution and selects at most $O(nD)$ pairs. The terminal check costs $O(n)$ queries. The resulting bounds are $O(C\varepsilon^{-2}nD^8)$ queries and $O(C\varepsilon^{-2}nD^6)$ output pairs. Pairs selected at different levels are disjoint, because selected pairs are excluded from every later predicate. All selected pairs are found through positive counts, and their multiplicities and stored weights are exact. This proves the support and size claims. The argument was uniform for every fixed input and every earlier transcript, proving the final conditional guarantee as well. □

14.7 The interface used by the implicit LP

Corollary 14.7 (Public positive semidefinite additions)**.** *For any known positive semidefinite matrix H, the same output gives*

$$(1-\varepsilon)(H+L) \preceq H + L_Q \preceq (1+\varepsilon)(H+L).$$

If $H+L$ is positive definite, this provides an explicit positive definite spectral preconditioner on the successful event.

Proof. Add H to (33); the two extra errors are $\varepsilon H \succeq 0$. Positive definiteness follows because $1-\varepsilon > 0$. □

This handles the explicit rows in the normal matrices of the LP execution of Section 15. Duplicate hidden rows have identical pair formulas, so their total contribution is exactly $c_e w_e a_e a_e^\top$. For positive dyadic input weights, an output weight is $c_e 4^r w_e$, increasing its representation length by only $O(R + \log C)$ bits. No irrational arithmetic is needed for the graph construction: comparisons of $1/w_e$ can be made by comparing positive dyadics in the opposite order.

The aggregation primitive used alongside sparsification is also elementary, and is recorded here to make the oracle interface explicit.

Lemma 14.8 (Dyadic row aggregates)**.** *Suppose a public signed dyadic formula $q(v,u)$, for $u \ne v$, has nonzero binary digits only in a fixed set of B bit positions. All values*

$$y_v = \sum_{u \ne v} c_{uv} q(v,u)$$

can be computed exactly with $O(nB)$ outgoing-cut queries. Any scalar sum $\sum_{u<v} c_{uv} q(u,v)$ has the same bound.

Proof. Use a sign and magnitude binary representation. For each vertex, bit position, and sign, form the public set of endpoints whose formula has that sign and that bit set in its magnitude. Its neighbor count is given by (34). Multiplying by the signed power of two and summing recovers y_v. For the scalar version, restrict the candidate set for row u to endpoints $v > u$, and sum the rows. There is no double counting. With $B = O(\log n)$, the cost is $\widetilde{O}(n)$. □

If an algorithm makes at most N adaptive spectral calls, it can give each call failure budget ζ/N. Each new pair formula is fixed conditional on the preceding transcript, and each call uses fresh coins. The adaptive union bound therefore gives simultaneous correctness with probability at least $1 - \zeta$, without an assumption that the calls' success events are independent.

This proves the graph and aggregation interfaces used in the body of the paper. Their correctness is independent of the numerical LP analysis.

15 The repaired stable implicit LP execution

The purpose of this section is to discharge the LP interface, rather than to assume an unspecified stable solver. We give the full numerical execution and its query implementation. The method is the weighted path following of Lee and Sidford [LS20]: the trigonometric barrier for box constraints [LS20, Section 3.1], the mixed norm and centrality measure [LS20, Definition 11], and ℓ_p Lewis weights [LS20, Definition 20] with $p = 1 - O(1/\log M)$ are theirs. The parameter repair follows the audit [LSAudit26], using the current sharper slack floor. Independently, Bai, Dai, and Peng [BDP26] analyze a different Lee–Sidford method under fixed-point arithmetic.

15.1 Statement and proof boundary

Consider the finite-box LP

$$\min c^\top x \quad \text{subject to} \quad A^\top x = b, \qquad \ell \le x \le u, \qquad A \in \mathbb{R}^{M \times r}. \tag{41}$$

Assume the following explicit structure: constant-multiplicity hidden incidence rows, identical public pair formulas on duplicate copies, and $O(r)$ explicit rows including $+I_r$. All row norms are at most $\sqrt{2}$, and the input is dyadic with polynomial numerical bounds. Remove any zero explicit row and optimize its variable separately. Henceforth all rows are nonzero.

Let $U \ge 2$ bound the box widths, $\|c\|_\infty$, the absolute input coordinates, and the reciprocal slack of the supplied strictly feasible point x^0. Let $a_0 > 0$ be a public lower bound on the norm of every row of A, available from the explicit input bit bound and the integral hidden rows. Write

$$\mathcal{N} = 2 + M + U + \varepsilon^{-1} + a_0^{-1}.$$

We may assume $0 < \varepsilon \le 1$, by solving to accuracy $\min\{\varepsilon, 1\}$ if needed. The assumptions imply $\mathcal{N} = r^{O(1)}$.

Theorem 15.1 (Stable implicit LP implementation)**.** *There is a specified modification of the weighted path following of [LS20] that, conditional on valid oracle outputs, returns an exactly feasible, ε-suboptimal point of (41). Every primal iterate is exactly feasible and strictly interior. Fixed precision $\mathcal{N}^{-O(1)}$ suffices for all scalar evaluations and normal solves. It uses $\widetilde{O}(\sqrt{r})$ batches, each consisting of polylogarithmically many row reductions, leverage computations, normal solves, and mixed-ball support computations.*

Using the weighted spectral sparsifier and dyadic aggregation primitives of Section 14, the implementation uses $\widetilde{O}(r^{3/2})$ outgoing-cut queries with high probability. Hidden coordinates retain public

pair formulas, common to duplicate copies. Only oracle queries are charged; offline computation is unrestricted.

The proof below is a replacement execution, with the sensitivity repair of [LSAudit26] and all changed envelopes substituted. It is a mathematical proof of the modified execution, not an invocation of the disputed inexact-solver interface. Section 14 supplies its actual graph-query primitives.

15.2 Parameters and local facts

Use the finite-interval barrier

$$\phi_i(x_i) = -\log\cos\left(\frac{\pi(x_i - (\ell_i + u_i)/2)}{u_i - \ell_i}\right), \quad S(x) = \operatorname{diag}\sqrt{\phi''(x)}, \quad A_x = S(x)^{-1}A.$$

For $s_i(x) = \min\{x_i - \ell_i, u_i - x_i\}$, direct calculation gives

$$|\phi_i'| \le \min\{S_i, s_i^{-1}\}, \qquad U^{-1} \le S_i \le 2/s_i, \qquad S_i s_i \ge 1, \qquad \phi_i'(x_i)(v_i - x_i) \le 1 \quad (\ell_i \le v_i \le u_i). \tag{42}$$

The usual self-concordance bounds [NN94][LS20, Lemmas 8–9] control Hessian changes over a step whose normalized coordinate magnitude is less than one.

Choose a public power of two $\overline{M}$ with $M \le \overline{M} < 2M$, and a power of two L_0 at least $\max\{2, \lceil \log_2(4\overline{M})\rceil\}$, within a factor two of this lower bound, and put

$$p = 1 - L_0^{-1}, \qquad k = 2L_0, \qquad C_N = 48k, \qquad g(x) = w_p(A_x) + c_0, \qquad c_0 = 1/(2\overline{M}).$$

Here w_p is the unregularized Lewis-weight vector, and the scalar regularization is added to each coordinate. For positive w, define

$$N_w(v) = \|v\|_\infty + C_N\|v\|_w, \qquad \delta_t(x, w) = \min_\nu N_w\left(\frac{tc + w\phi'(x) - A\nu}{w\sqrt{\phi''(x)}}\right).$$

Here $\|v\|_w = (\sum_i w_i v_i^2)^{1/2}$ is the weighted Euclidean norm. All division in row space is coordinatewise.

The following local facts hold for these repaired parameters. The sensitivity and projection proof, and the needed corrections to the Lewis-movement argument, are given immediately below.

$$\sum_i g_i = r + Mc_0 \le r + 1/2, \quad c_0 \le g_i \le 1 + c_0, \tag{43}$$

$$\delta_t(x, w) \le N_w(Sh) \le \gamma\delta_t(x, w), \quad \gamma = 1 + (16k)^{-1}, \tag{44}$$

$$N_{g(x)}(\log g(x + d) - \log g(x)) \le (1 - k^{-1} + 4v)v, \quad v = N_{g(x)}(S(x)d) \le 1/10, \tag{45}$$

$$\delta_t(x, \widetilde{w}) \le (1 + 4a)(\delta_t(x, w) + a), \quad a = N_w(\log\widetilde{w} - \log w) \le 1/10. \tag{46}$$

Equation (44) requires $\frac{4}{5}g(x) \le w \le \frac{5}{4}g(x)$, and h is the exact feasible Newton direction. Equation (45) requires the segment to be interior. Here $L_0 - 1 \ge \log_2(2M)$ and the regularized sensitivity is at most 16. The projection bound uses the sharper Pythagorean estimate; consistency is valid for every nonnegative regularizer.

Lemma 15.2 (Small-regularizer sensitivity)**.** *Let $B \in \mathbb{R}^{M\times r}$ have full column rank and no zero rows, and let $w = w_p(B)$, $0 < p \le 1$. If $0 < \epsilon < 1$ and $0 \le c \le \epsilon/M$, then $g = w + c\mathbf{1}$ satisfies*

$$\frac{b_i^\top(B^\top G^{-1}B)^{-1}b_i}{g_i^2} \le \frac{2}{1 - \epsilon}(M/\epsilon)^{2/p-2}.$$

In particular the parameters above give sensitivity at most 16.

Proof. Put $a = 2/p - 2 \geq 0$ and $H = B^\top W^{1-2/p} B$. The Lewis identity says $b_i^\top H^{-1} b_i = w_i^{2/p}$; also $0 < w_i \leq 1$ and $\sum_i w_i = r$. Call a row light if $w_i \leq \epsilon/M$. For its part H_{light},

$$\operatorname{tr}(H^{-1} H_{\text{light}}) = \sum_{\substack{i \\ w_i \leq \epsilon/M}} w_i \leq \epsilon.$$

Since the normalized matrix is positive semidefinite, it is bounded by its trace times the identity. Thus $H_{\text{light}} \preceq \epsilon H$ and $H \preceq (1-\epsilon)^{-1} H_{\text{heavy}}$. On a heavy row $g_i \leq 2w_i$, so

$$w_i^{1-2/p} = w_i^{-a}/w_i \leq 2(M/\epsilon)^a / g_i.$$

Consequently $H \preceq [2/(1-\epsilon)](M/\epsilon)^a B^\top G^{-1} B$. Invert this inequality and evaluate its quadratic form at b_i. The remaining factor $w_i^{2/p}/g_i^2$ is at most $w_i^{2/p-2} \leq 1$. With $\epsilon = 1/2$, $a = 2/(L_0 - 1)$ and $(2M)^a \leq 4$, the bound is 16. □

Lemma 15.3 (Pythagorean projection estimate)**.** *Let P be the w-orthogonal projection onto feasible normalized directions and $Q = I - P$. If $\|Qy\|_\infty \leq \sqrt{q}\,\|Qy\|_w$ for every y, then*

$$\|P\|_{N_w \to N_w} \leq \sqrt{1 + q/C_N^2}.$$

If $\frac{4}{5} g \leq w \leq \frac{5}{4} g$, the preceding lemma gives $q \leq 32$ and therefore this norm is at most $1 + 1/(16k)$.

Proof. Pythagoras and the triangle inequality give, with $t = \|Qy\|_w$,

$$N_w(Py) \leq \|y\|_\infty + \sqrt{q}\, t + C_N \sqrt{\|y\|_w^2 - t^2} \leq \|y\|_\infty + \sqrt{q + C_N^2}\, \|y\|_w.$$

For $B = A_x$, the diagonal quadratic forms defining q are $w_i^{-2} b_i^\top (B^\top W^{-1} B)^{-1} b_i$. Replacing g by a vector in the displayed window increases them by at most $(5/4)^3 < 2$, hence $q \leq 32$. Finally, $\sqrt{1 + 32/(48k)^2} \leq 1 + 1/(144k^2) \leq 1 + 1/(16k)$. The exact normalized Newton direction is minus P applied to the normalized gradient. Projection kills its row-space component, so taking the minimum over multipliers proves (44). □

Existence, differentiability, and weight movement. For completeness, the Lewis map in Lemma 15.9 is a contraction on log coordinates. The complete space $\mathbb{R}^M$ therefore has a unique fixed point. Its positive weights satisfy the leverage identity, which implies the size facts in (43). This also avoids an unsupported restriction-to-a-box argument for existence. Under a log row scaling $B \mapsto \operatorname{diag}(e^s) B$, the map is smooth in both s and the log weights. Its derivative in the latter variable has norm at most $1 - p/2 < 1$, so the implicit function theorem applies to the fixed point equation. Uniqueness identifies this local differentiable solution with the global weights.

Let Π be the orthogonal projection associated with $W^{1/2-1/p} B$, and write $\Lambda = W - \Pi \circ \Pi$, where $\circ$ is entrywise product. Differentiating the leverage identity gives the log-scaling Jacobian

$$K = 2\big(W + (2/p - 1)\Lambda\big)^{-1} \Lambda.$$

The projection-Jacobian inequalities of [LS20, Lemmas 24–25] give $\|Kh\|_w \leq p\|h\|_w$ and $\|Kh - ph\|_\infty \leq p\|h\|_w$ for $p \leq 1$. Regularization differentiates to $(w/g)Kh$. Because $g \geq w$, its weighted norm is no larger than $\|Kh\|_w$, and its infinity norm is no larger than $\|Kh\|_\infty$. It follows that

$$N_g((w/g)Kh) \leq p(1 + 1/C_N) N_g(h) \leq (1 - 1/k) N_g(h).$$

For the barrier scaling, the derivative of $\log(S^{-1})$ in a direction d has coordinates $\rho_i S_i d_i$ with $|\rho_i| \le 1$. Along an interior segment, self-concordance bounds the normalized step by its initial value divided by $1 - v$. If u bounds the accumulated infinity-norm change of $\log g$, changing the mixed norm at both ends of this operator estimate costs e^u, since each norm comparison costs $e^{u/2}$. The continuity bootstrap closes because, for $0 < v \le 1/10$ and $0 < a \le 1$,

$$\frac{e^{(1-a+4v)v}(1-a)}{1-v} < 1 - a + 4v.$$

For example $e^{1.4v} \le 1 + 1.6v$ and $(1-v)^{-1} \le 1 + 1.12v$ imply an upper bound $(1-a)(1+2.9v) < 1 - a + 4v$. Integrating with $a = 1/k$ proves (45). This uses the sharp norm comparison twice, as required; no single-factor operator transfer is assumed.

For (46), put $\ell = \log \widetilde{w} - \log w$. Using the same multiplier, the new normalized residual is the old one multiplied coordinatewise by $e^{-\ell}$ plus $(1 - e^{-\ell})\phi'/S$. Since $|\phi'/S| \le 1$ and $\|\ell\|_\infty \le a$, its mixed norm is at most $e^{3a/2}(\delta_t(x,w) + a) \le (1 + 4a)(\delta_t(x,w) + a)$. Taking the infimum proves the assertion. The infimum is attained because it is distance to a closed finite-dimensional subspace in a norm. The frozen center exists because the interval barrier diverges at the boundary of a bounded feasible box, and it is unique by strict convexity. The exact Newton direction exists uniquely on the feasible tangent space because its Hessian is positive definite. These facts justify all minimizers used below.

The following parameters are fixed throughout path following:

$$\begin{aligned}
&T_0 = 1 + 2C_N\lceil\sqrt{r}\rceil, \qquad \xi = (8k)^{-1}, \qquad && K = (2^{20}k)^{-1},\\
&B_0 = 12MT_0/\xi, \qquad L_B = \text{a power of two in } [\lceil\log_2 B_0\rceil, 2\lceil\log_2 B_0\rceil), &&\\
&\Delta = \frac{\xi K}{16 L_B}, \qquad \mu = \frac{\xi}{12\Delta}, && C_* = \left(1 - \frac{3}{4k}\right)\Delta.
\end{aligned}$$

Thus $\Delta = \log^{-O(1)} M$, $\Delta \le (2^{16}k^2)^{-1}$, and $\mu^{-1}\log B_0 \le K$. Define

$$\Psi(v) = \sum_i (e^{\mu v_i} + e^{-\mu v_i}).$$

The maintained invariants, with $X = \log w$, are

$$A^\top x = b, \qquad \ell < x < u, \qquad \delta_t(x,w) \le \Delta, \qquad \Psi(X - \log g(x)) \le B_0. \tag{47}$$

In particular $\|X - \log g(x)\|_\infty \le K$, $1/(4\overline{M}) \le w_i \le 4$, $\sum_i w_i \le 4r$, and $N_w(v) \le T_0\|v\|_\infty$.

15.3 Absolute primal errors and robust chasing

Lemma 15.4 (Perturbed feasible Newton step)**.** *Suppose the weight comparison in (44) holds. Let $d = h + e$, with $A^\top e = 0$, and suppose $N_w(Se) \le \eta$. If $a = \gamma\delta_t(x,w) + \eta \le 1/10$, then $x + d$ is strictly interior and exactly feasible, and*

$$\delta_t(x + d, w) \le 8\delta_t(x,w)^2 + 4\eta. \tag{48}$$

Proof. Put $z = Sd$, so $N_w(z) \le a$. There is a multiplier π with $tc + W\phi'(x) + WS^2 h = A\pi$. Use that multiplier in the definition of the new centrality. The remaining residual is

$$W(S^2 e + \phi'(x + d) - \phi'(x) - S^2 d).$$

Self-concordance gives $S_i(x+d) \geq (1-a)S_i(x)$, and integration of the Hessian variation gives

$$\frac{|\phi_i'(x_i+d_i) - \phi_i'(x_i) - S_i^2 d_i|}{S_i(x+d)} \leq \frac{|z_i|^2}{(1-a)^2}.$$

Since $N_w(z^2) \leq \|z\|_\infty N_w(z)$, the new centrality is at most $\eta/(1-a) + a^2/(1-a)^2$. For $a \leq 1/10$, $\gamma \leq 17/16$, this is at most $8\delta^2 + 4\eta$. Interiority follows from the same normalized-step bound. □

Lemma 15.5 (Chasing with an approximate support point)**.** *Let $\mathcal{U}$ be an unconditional, symmetric convex set containing $[-C_*/T_0, C_*/T_0]^M$ and contained in $[-C_*, C_*]^M$. Suppose the target moves from y to $y' = y+u$, with $u \in \mathcal{U}$, and $\|z - y'\|_\infty \leq \Delta$. Set $q = \nabla\Psi(X-z)$. Let $v \in \mathcal{U}$ be sign-aligned with q, and satisfy*

$$\langle q, v\rangle \geq h_{\mathcal{U}}(q) - \sigma, \qquad \sigma \leq \xi/1000.$$

For $X' = X - (1+\xi)v$,

$$\Psi(X' - y') \leq (1-\lambda)\Psi(X-y) + \frac{\xi M C_*}{3\Delta} + 2\sigma, \qquad \lambda = \frac{\xi^2 C_*}{30\Delta T_0}. \tag{49}$$

In particular $\Psi(X-y) \leq B_0$ implies $\Psi(X'-y') \leq B_0$.

Proof. For $f(a) = e^{\mu a} + e^{-\mu a}$, $|a-b| \leq d$, and $\mu d \leq 1/5$,

$$e^{-\mu d}|f'(b)| - \mu \leq \operatorname{sign}(b) f'(a) \leq e^{\mu d}|f'(b)| + \mu.$$

This follows by writing the derivative as $2\mu\sinh(\mu a)$; the additive term is bounded by $2\mu\sinh(\mu d) \leq \mu$. Summing against a sign-aligned vector proves the corresponding inner-product comparison. Unconditionality ensures that a support maximizer can be chosen sign-aligned, and hence also gives

$$h_{\mathcal{U}}(\nabla\Psi(a)) \leq e^{\mu d} h_{\mathcal{U}}(\nabla\Psi(b)) + \mu M C_*.$$

Apply the mean value theorem first to the target move and then to the player move. Their comparison distances are at most Δ and 3Δ, respectively, and the observed point differs from the old target by at most 2Δ. Writing $H = h_{\mathcal{U}}(\nabla\Psi(X-y))$ gives

$$\begin{aligned}\Psi(X'-y') &\leq \Psi(X-y) - ((1+\xi)e^{-5\mu\Delta} - e^{\mu\Delta})H + (3+2\xi)\mu M C_* + (1+\xi)\sigma \\ &\leq \Psi(X-y) - \frac{2\xi}{5}H + \frac{7\xi M C_*}{24\Delta} + 2\sigma.\end{aligned}$$

For the second inequality use $\xi \leq 1/8$, $\mu\Delta = \xi/12$, and $(1+\xi)e^{-5\xi/12} - e^{\xi/12} \geq 2\xi/5$. For example, $e^{-v} \geq 1 - v$ and $e^v \leq 1 + v + v^2$ in this range prove the latter inequality.

The contained cube and $\|\nabla\Psi(a)\|_1 \geq \mu(\Psi(a) - 2M)$ imply $H \geq (C_*/T_0)\mu(\Psi(X-y) - 2M)$. Substitution gives (49). Finally $\lambda B_0 = 2\xi M C_*/(5\Delta)$, whereas $2\sigma \leq \xi M C_*/(30\Delta)$, since $C_* \geq \Delta/4$. The additive terms are therefore strictly smaller than λB_0. □

Proposition 15.6 (One robust path batch)**.** *Suppose (47) holds and $\eta \leq \Delta/(2^{16}k)$. Perform these operations:*

1. *Form an exactly feasible direction d satisfying $N_w(S(d-h)) \leq \eta$, and put $x' = x + d$.*
2. *Obtain a dyadic z with $\|z - \log g(x')\|_\infty \leq \Delta$.*
3. *Apply Lemma 15.5 with $\mathcal{U} = \{v : N_w(v) \leq C_*\}$, and set $w' = \exp(X - (1+\xi)v)$.*

Then the potential invariant holds at (x', w'), and

$$\delta_t(x', w') \le \left(1 - \frac{1}{2k}\right)\Delta. \tag{50}$$

No evaluation of δ_t is required by these operations.

Proof. Let $d = h + e$. Since $N_g \le e^{K/2} N_w$,

$$v_g = N_g(Sd) \le e^{K/2}(\gamma\Delta + \eta) \le (1 + (15k)^{-1})\Delta.$$

Equation (45), followed by the reverse norm comparison, shows

$$N_w(\log g(x') - \log g(x)) \le \left(1 - \frac{7}{8k}\right)\Delta < C_*. \tag{51}$$

For explicit bounds, use $e^{K/2}(1+(15k)^{-1}) \le 1+(14k)^{-1}$ and $4v_g \le (10^4 k)^{-1}$. The resulting product is at most $1 - 7/(8k)$. Thus the target move belongs to $\mathcal{U}$. Also $N_w(v) \le T_0\|v\|_\infty$, so the geometric hypotheses of Lemma 15.5 hold. It proves the potential invariant.

The log-weight step has mixed norm

$$a_w \le (1 + \xi)C_* \le (1 - 5/(8k))\Delta.$$

By Lemma 15.4 and (46),

$$\begin{aligned}\delta_t(x', w') &\le (1 + 4\Delta)(8\Delta^2 + 4\eta + (1 - 5/(8k))\Delta) \\ &\le (1 - 1/(2k))\Delta.\end{aligned}$$

Here $8\Delta + 4\eta/\Delta \le (4096k)^{-1}$ and $4\Delta \le (16384k)^{-1}$, leaving more than the stated margin. □

For a cost parameter change $|t'/t - 1| \le \alpha$, scaling the multiplier in the centrality definition and using (42) gives

$$\delta_{t'}(x, w) \le (1 + \alpha)\delta_t(x, w) + \alpha T_0. \tag{52}$$

Choose a dyadic α between $\Delta/(32kT_0)$ and $\Delta/(16kT_0)$. Following each batch by a move toward the desired endpoint with relative change at most α preserves (47). It takes $O((kT_0/\Delta)|\log(t_{\text{end}}/t_{\text{start}})| + 1)$ batches to traverse a phase. This is $\widetilde{O}(\sqrt{r})$.

15.4 A noncircular polynomial slack bound

Lemma 15.7 (Frozen-weight center)**.** *Suppose $\|\log w - \log g(x)\|_\infty \le 1/20$ and $\delta_t(x, w) \le \Delta$. The exact center*

$$\bar{x} = \arg\min_{A^\top y = b} \left(tc^\top y + \sum_i w_i \phi_i(y_i)\right)$$

satisfies

$$\tfrac{1}{2}s_i(x) \le s_i(\bar{x}) \le 2s_i(x), \qquad \|x - \bar{x}\|_\infty \le 4U\delta_t(x, w). \tag{53}$$

Proof. This is an analytic comparison, not an additional algorithmic phase. Run exact Newton steps with t, w frozen. As long as the weights remain within factors $4/5, 5/4$ of their targets, Lemma 15.4 gives $\delta_{j+1} \le 8\delta_j^2 \le \delta_j/2$. The normalized step has norm at most $\gamma\delta_j \le 2\delta_j$. Equation (45) then bounds the change of $\log g$ in infinity norm by $4\delta_j$. Its total change is at most $8\delta_0$, which closes the weight-comparison induction.

By $S_i s_i \ge 1$, each slack changes by a relative amount at most $2\delta_j$. The product of these factors lies between $1/2$ and 2. The steps converge to an interior feasible limit, their stationarity residual tends to zero, and strict convexity identifies the limit with $\bar{x}$. Finally $S_i \ge 1/U$ gives $\sum_j \|h_j\|_\infty \le 2U\sum_j \delta_j \le 4U\delta_0$. □

Lemma 15.8 (Center slack from an interior reference)**.** *Suppose $w_i \geq w_{\min} > 0$, and let the reference x^0 have minimum slack s_0. At a weighted center for cost vector c_*,*

$$s_i(\bar{x}) \geq \min\left\{\frac{s_0}{2}, \frac{w_{\min} s_0}{4(tMU\|c_*\|_\infty + \sum_j w_j)}\right\}.$$

Proof. Stationarity and feasibility imply

$$\sum_j w_j \phi_j'(\bar{x}_j)(x_j^0 - \bar{x}_j) = -tc_*^\top(x^0 - \bar{x}).$$

Each summand is at most w_j. If the lower slack of coordinate i is $s < s_0/2$, then $x_i^0 - \bar{x}_i \geq s_0/2$. The trigonometric barrier obeys $-\phi_i'(\bar{x}_i) \geq 1/(2s)$ for $s \leq (u_i - \ell_i)/4$. Hence the negative of this summand is at least $w_{\min} s_0/(4s)$ and at most $tMU\|c_*\|_\infty + \sum_j w_j$. The upper endpoint is identical. □

Throughout the algorithm below, $t \leq 16r/\varepsilon$, $\|c_*\|_\infty \leq 4U$, $w_i \geq 1/(4\overline{M})$, and $\sum_i w_i \leq 4r$. Applying these two lemmas with $s_0 \geq 1/U$ gives the uniform conservative bound

$$s_i(x) \geq s_* := \frac{\varepsilon}{2^{12} r \overline{M}^2 U^3}. \tag{54}$$

It also applies immediately after a primal update, before the weight update: centrality has decreased and the old weight differs from the new target by at most $K + C_* < 1/20$. This proves the required envelope for the modified execution itself.

15.5 Stable evaluation of the primitives

15.5.1 Lewis weights: a contraction in log coordinates

Lewis weights are due to Lewis [Lew78]. The contraction below is Lemma 3.2 of Cohen and Peng [CP15], restated in log coordinates with the error recurrence used here.

Lemma 15.9 (Stable Lewis fixed point)**.** *For a full-rank matrix B without zero rows and $1/2 \leq p < 1$, define, for positive v,*

$$\mathcal{T}_B(v)_i = \left(b_i^\top (B^\top V^{1-2/p} B)^{-1} b_i\right)^{p/2}.$$

Then

$$\|\log \mathcal{T}_B(v) - \log \mathcal{T}_B(u)\|_\infty \leq (1 - p/2)\|\log v - \log u\|_\infty. \tag{55}$$

Its fixed point is $w_p(B)$. A per-iteration log error at most ζ gives the error recurrence $E_{j+1} \leq (1 - p/2)E_j + \zeta$.

Proof. Put $a = 1 - 2/p < 0$ and $E = \|\log v - \log u\|_\infty$. The two normal matrices are within Loewner factors $e^{\pm|a|E}$. The same is true of their inverse quadratic forms. Taking the power $p/2$ proves (55), since $p|a|/2 = 1 - p/2 \leq 3/4$. The Lewis fixed-point identity identifies the fixed point. The last assertion is the triangle inequality. □

Implement an iteration using simultaneous leverage estimates:

$$\log \mathcal{T}_B(v)_i = \frac{p}{2} \log \sigma_i(V^{1/2-1/p} B) + (1 - p/2) \log v_i.$$

Freeze the output log coordinates on a dyadic mesh. For example, use per-iteration log error at most $\Delta/128$, and take enough iterations that $(3/4)^j \log(1/q_*) \leq \Delta/32$, with q_* defined below. The

final log error is at most $\Delta/16$. Leverage accuracy $O(\Delta)$ and smaller rounding error suffice. Adding c_0 cannot increase the log error, so a dyadic estimate of $\log g(x)$ to accuracy Δ follows.

Here is the spectral implementation of a leverage call. For $B = A_x$, the row conductances in $B^\top V^{1-2/p}B$ are $\omega_i = v_i^{1-2/p}/S_i^2$. The restart and slack bounds below put every actual ω_i between public inverse-polynomial and polynomial bounds. Freeze them to positive dyadics with coordinatewise relative error θ, and call the spectral routine with error θ, where θ is a sufficiently small fixed multiple of Δ. The hidden part is a Laplacian because all copies of a pair share the same formula; keep the explicit PSD contribution exactly at its frozen value. The two relative errors compose to $1 \pm O(\theta)$. Inversion gives the corresponding sandwich for every row quadratic form, so its logarithmic error is $O(\theta)$. The row scaling used to recover the leverage contributes another $O(\theta)$. All inverse quadratic forms of the explicit output matrix are evaluated offline, on every potential incidence row, without adjacency queries. A sufficiently small dyadic log-rounding mesh then gives the per-iteration error used above. Inverse-polynomial conductance floors permit relative freezing with $O(\log \mathcal{N})$ bits. Thus the leverage calls use inverse-polylogarithmic spectral accuracy, while Newton solves need only constant spectral accuracy followed by refinement.

There is a polynomial initial envelope for restarting this iteration from $v_i = 1$ at every call. If j minimizes $w_p(B)_i$, the Lewis identity and $a < 0$ imply

$$B^\top W_p^a B \preceq (w_p)_j^a B^\top B, \qquad (w_p)_j \geq b_j^\top (B^\top B)^{-1} b_j.$$

For $B = A_x$, (42) and (54) therefore give

$$1 \geq (w_p)_i \geq q_* := \frac{a_0^2 s_*^2}{8MU^2}. \tag{56}$$

The initial log error is at most $\log(1/q_*)$. The contraction keeps all intermediate weights between q_*^2/e and e/q_*. All scaled normal matrices are consequently polynomially conditioned. Using $O(\log \mathcal{N})$ iterations is more than sufficient. This also supplies initialization; no warm-start or homotopy stability claim is needed.

15.5.2 Newton solves and the exact star repair

For $H = WS^2$, $D = H^{-1}$, $L = A^\top DA$, and $q = tc_* + W\phi'(x)$, the exact direction is

$$h = -D(q - A\pi), \qquad L\pi = A^\top Dq.$$

The established envelope gives

$$\frac{s_*^2}{16} I \preceq L \preceq 8M\overline{M}U^2 I, \qquad \|D\|_2 \leq 4\overline{M}U^2, \qquad \|q\|_\infty \leq \frac{64rU}{\varepsilon} + \frac{4}{s_*}. \tag{57}$$

The lower bound uses the explicit $+I_r$ block.

Freeze D to coordinatewise relative error β and q to absolute coordinate error β. The lower bound $D_i \geq s_*^2/16$ permits a fixed dyadic grid of width at most $\beta s_*^2/32$ for D; a grid of width at most $\beta/2$ suffices for q. The resulting matrix is a hidden dyadic Laplacian plus an explicit PSD matrix. Construct a constant-quality spectral preconditioner using Section 14. Starting from zero, preconditioned Richardson iteration for the frozen system contracts energy error by at most $1/2$ per step (decrease the fixed sparsifier error if necessary). If each computed step has energy error at most τ, then

$$\|\pi_j - \widehat{\pi}\|_{\widehat{L}} \leq 2^{-j} \|\widehat{\pi}\|_{\widehat{L}} + 2\tau.$$

This follows by induction from the one-step triangle inequality. It is an absolute guarantee, valid also when the right-hand side is zero. The public polynomial bounds below bound the initial norm. Hence $O(\log \mathcal{N})$ steps and fixed inverse-polynomial τ make the absolute Euclidean potential error at most β. More explicitly, if $\underline{\lambda} > 0$ and $\overline{\lambda}$ are the public frozen eigenvalue bounds and P_π is the solution bound below, it suffices to choose

$$2^{-j}\sqrt{\overline{\lambda}}\,P_\pi \le \tfrac{1}{2}\beta\sqrt{\underline{\lambda}}, \qquad \tau \le \tfrac{1}{4}\beta\sqrt{\underline{\lambda}}.$$

All these choices are fixed inverse-polynomial tolerances. Coordinate rounding of width δ contributes at most $\sqrt{\lambda_{\max}(\widehat{L})}\sqrt{r}\,\delta$ to τ. Each matrix product is an exact dyadic aggregate of the public pair formula $\widehat{D}_{uv}(\pi_u - \pi_v)$; the explicit contribution is free. The energy recurrence and eigenvalue floor keep all represented potentials polynomially bounded, so these products use $O(\log \mathcal{N})$ bit positions. All raw direction coordinates are formed exactly from the frozen dyadics.

Let $R_* : \mathbb{R}^r \to \mathbb{R}^M$ insert a vector into the explicit $+I_r$ row coordinates and be zero on all other rows. Thus $A^\top R_* = I_r$ and $\|R_* z\|_\infty = \|z\|_\infty$. This specific right inverse is used for the repair below; it changes no hidden coordinate.

Lemma 15.10 (Absolute direction accuracy)**.** *There is a public fixed polynomial $P(\mathcal{N})$ such that the resulting raw direction $\widetilde{h}$, repaired as*

$$d = \widetilde{h} - R_* A^\top \widetilde{h}, \qquad A^\top R_* = I_r,$$

satisfies $A^\top d = 0$ and $N_w(S(d-h)) \le P(\mathcal{N})\beta$.

Proof. For precision at most $1/4$, (57) holds for the frozen system with a factor two change. Matrix and right-hand-side perturbations have absolute norm $\mathrm{poly}(\mathcal{N})\beta$. The identities

$$\widehat{\pi} - \pi = \widehat{L}^{-1}\big((\widehat{b} - b) - (\widehat{L} - L)\pi\big), \qquad \|\pi\|_2 \le \|L^{-1}\|_2 \|b\|_2,$$

therefore bound the potential error absolutely by a fixed polynomial times β. The absolute Richardson bound above gives the same fixed-polynomial potential error, because the frozen solution and inverse eigenvalue are polynomially bounded. Substitution in $-D(q - A\pi)$ proves $\|\widetilde{h} - h\|_\infty \le P_1(\mathcal{N})\beta$. Only a fixed number of matrix-product and inverse-norm bounds is being used here, not a composition over path iterations.

The repair residual is an exact dyadic aggregate. Since $A^\top h = 0$,

$$d - h = (I - R_* A^\top)(\widetilde{h} - h), \qquad N_w(S(d-h)) \le \frac{2T_0}{s_*}(1 + \sqrt{2}M)\|\widetilde{h} - h\|_\infty.$$

This gives the asserted polynomial and exact feasibility. No relative accuracy with respect to the original solution π is asserted; the argument remains valid when its right-hand side is zero. □

For clarity, an explicit polynomial can be read off without any condition-number convention. Put $d_+ = 4\overline{M}U^2$, $\ell_* = s_*^2/16$, $a = \sqrt{2M}$, and $Q = 64rU/\varepsilon + 4/s_*$. One may take

$$\begin{aligned}
P_\pi &= 4a\sqrt{M}\,d_+(Q+1)/\ell_*, \\
E_\pi &= \frac{2}{\ell_*}\big(a\sqrt{M}\,d_+(Q+2) + a^2 d_+ P_\pi\big) + 2a\sqrt{d_+/\ell_*}\,P_\pi + \sqrt{r}, \\
P_1 &= d_+(Q + \sqrt{2}P_\pi) + 2d_+(1 + \sqrt{2}E_\pi), \\
P &= \frac{2T_0}{s_*}(1 + \sqrt{2}M)P_1.
\end{aligned}$$

Replacing these expressions by larger integer upper bounds is harmless. Choose $\beta \le \min\{1/4, \eta/P\}$.

15.5.3 Support accuracy and finite representation

At a path batch, $q = \nabla\Psi(X - z)$ obeys $\|q\|_\infty \le 2\mu B_0$: the observed argument differs from $X - \log g(x)$ by at most 2Δ. We give a stable scalar implementation, including the near-zero cases. Sign alignment is implementable exactly: X, z are dyadic, and the sign of the gradient is the sign of $X - z$.

Lemma 15.11 (Additive support oracle)**.** *Let $0 < \sigma < 1$, and suppose the objective signs are publicly decidable. Suppose the positive norm weights, their reciprocals, the objective magnitudes, the reciprocal radius, and σ^{-1} are bounded by a fixed polynomial in $\mathcal{N}$. A dyadic, sign-aligned $v \in \{v : N_w(v) \le C_*\}$ with*

$$\langle q, v\rangle \ge h_{\{N_w \le C_*\}}(q) - \sigma$$

can be computed using $\log^{O(1)}\mathcal{N}$ certified row aggregates, each to fixed inverse-polynomial accuracy.

Proof. Set

$$z = \frac{C_N}{C_*} W^{1/2} v, \qquad \kappa_i = C_N\sqrt{w_i}, \qquad a_i = \frac{C_* q_i}{C_N\sqrt{w_i}}.$$

The normalized support problem is $\max a^\top z$ subject to $\|z\|_2 + \|\kappa^{-1}z\|_\infty \le 1$. All transformed data have polynomial upper bounds, with polynomial lower bounds for positive κ_i.

Choose $\zeta = \sigma/1000$, and regularize the objective to $f_\rho(z) = a^\top z - \rho\|z\|_2^2$, where $\rho = \zeta$. The optimum decreases by at most ρ. For $\tau \in [0, 1]$, let

$$G_\rho(\tau) = \max\{f_\rho(z) : \|z\|_2 \le 1 - \tau,\ |z_i| \le \tau\kappa_i\}.$$

This function is concave: interpolate feasible pairs (τ, z) and use concavity of f_ρ. Scaling a feasible vector at one parameter to be feasible at another shows that it is Lipschitz with constant at most $L = \max\{\|a\|_2, \sum_i |a_i|\kappa_i\}$. The regularization term only improves these scaling inequalities.

Let $L_+ \ge \max\{1, L\}$ and $A_+ \ge \max\{1, \|a\|_2\}$ be public polynomial upper bounds. Restrict τ to $[h, 1 - h]$, where $h = \min\{1/4, \zeta/(8L_+)\}$; this loses at most $\zeta/8$. For a fixed τ, write $r_\tau = 1 - \tau$. The optimizer has the form

$$z_i(\lambda) = \operatorname{sign}(a_i)\min\left\{\tau\kappa_i, \frac{|a_i|}{2(\rho + \lambda)}\right\}, \qquad \lambda \ge 0. \tag{58}$$

Use $\lambda = 0$ if it meets the Euclidean constraint; otherwise bisect the monotone norm to find $\|z(\lambda)\|_2 = r_\tau$. The search interval $[0, \Lambda]$, $\Lambda = A_+/h$, suffices. Crucially, every denominator is at least 2ρ. The map $\lambda \mapsto z(\lambda)$ is globally Lipschitz in Euclidean norm with constant at most $A_+/(2\rho^2)$.

Here is a stopping and accuracy certificate. If a norm interval overlaps r_τ, evaluate it to width δ_n and conservatively rescale the vector into the ball. The scalar dual value

$$D(\lambda) = f_\rho(z(\lambda)) + \lambda(r_\tau^2 - \|z(\lambda)\|_2^2)$$

is an upper bound on $G_\rho(\tau)$. For a feasible vector with norm within δ_n of r_τ, its gap from this upper bound is at most $2\Lambda\delta_n$. For an infeasible vector within that norm error, scaling to radius r_τ loses at most $A_+\delta_n$ in objective, and the dual correction is nonpositive. The case $\lambda = 0$ needs no complementarity estimate. Alternatively, stop when the multiplier bracket has width $O(\rho^2\delta_n/A_+)$, using the Lipschitz bound above. Taking $\delta_n \le \zeta/[100(A_+ + \Lambda + 1)]$ thus gives an additive $O(\zeta)$ value interval and a feasible vector. All the required norm and objective sums are row aggregates of the coordinate formula (58). Their interval widths and the further rounding losses can be made smaller by a fixed polynomial factor. The number of bisection steps is $O(\log\mathcal{N})$.

The aggregates used here are squared norms and objectives. There is no exact hidden Euclidean-norm oracle. Since $r_\tau \geq h$, choose their absolute error δ_2 to satisfy

$$\delta_2 \leq \min\{3h^2/4,\ h\delta_n/4\}.$$

On an overlapping squared-norm interval this converts to the required norm error using $|a-b| = |a^2-b^2|/(a+b)$ and the lower radius h. Rescale using a certified upper norm bound; the result is feasible. The loss from changing squared radius by δ_2 is at most $A_+\delta_2/(2h)$, and every dual/objective sum is evaluated to a still smaller fixed fraction of ζ. These are explicit inverse-polynomial thresholds. The normalization scalars and every coordinate formula can be evaluated with certified intervals; division denominators are bounded below by h or ρ. Thus scalar evaluation errors can be made smaller than these thresholds without extra kinds of oracle access.

Maximize G_ρ by trisection with certified value intervals. When the intervals separate, discard the worse exterior third. When intervals of width at most ζ overlap, retain the middle third and the best queried incumbent. Concavity bounds each discarded exterior third by its neighboring queried value plus 2ζ. Keeping the incumbent prevents accumulation of these losses. After $O(\log(L_+/\zeta))$ evaluations, the remaining interval has width at most ζ/L_+. Its Lipschitz bound finishes the optimization to additive $O(\zeta)$ accuracy.

Undo the diagonal scaling, and round each output magnitude downward on a dyadic grid. This preserves both feasibility and sign alignment; the objective loss is at most $\sum_i |q_i|$ times the grid width. Certified lower intervals for the magnitudes implement this operation without exact comparisons of analytic numbers. All bounds are polynomial and ρ, h are inverse polynomial. Taking the mesh widths small enough makes the total loss at most σ, with the stated number of aggregate queries. □

Apply this lemma with $\sigma = \xi/1000$. The radius C_* is fixed and inverse polylogarithmic, so no positive lower bound on the actual centrality is assumed.

All grid widths can be fixed before path following. To see this without assuming a hidden precision oracle, take public polynomial upper bounds for $M, U, \varepsilon^{-1}, a_0^{-1}, s_*^{-1}, q_*^{-1}, P$ and for the support data, reciprocals h^{-1}, ρ^{-1}, and interval tolerances just listed. Choose a common dyadic mesh below the minimum of the finitely many explicit accuracy thresholds divided by these polynomial bounds. Approximate analytic coordinate formulas to that mesh before every aggregate; summing over at most M rows multiplies absolute error by at most M. The Lewis restart contracts by $3/4$, Richardson contracts by $1/2$, and the support searches have logarithmic depth with an incumbent, so precision is never propagated through an uncontrolled long product of condition numbers. Rounded D remains positive by its established lower bound. Final support magnitudes are rounded toward zero from certified lower interval endpoints; feasibility and sign alignment are preserved and the loss is at most the mesh times $\sum_i |q_i|$. Choosing a smaller fixed mesh to absorb this last loss preserves all stated query bounds.

Store X as exact sums of dyadic support updates. Store x as exact sums of repaired dyadic directions. The update grids can be fixed in advance; addition does not multiply their denominators. Values entering bit-class aggregation have $\log^{O(1)} \mathcal{N}$ bits. Public parameters and analytic barrier circuits are evaluated only through the specified certified interfaces. In particular, an exact public path parameter need not itself be sent to a row-aggregation primitive.

Pair evaluators are made total by public domain guards and fallback values on invalid potential rows. For example, use a fallback before evaluating a barrier outside its interval, and clip intermediate log-weight representations to a larger polynomial envelope. The proved slack and weight bounds ensure that no actual row is changed by these guards on the successful execution. Their predicates use only the represented data, not an unqueried adjacency bit.

These guards also specify the execution after a failed spectral call. Fix all loop limits, query budgets, dyadic meshes, and polynomial magnitude envelopes in advance from the public parameters above. Before each aggregate, round and clip its public pair coefficients to the prescribed bit range; before a logarithm, inverse, barrier evaluation, or support division, test its public domain conditions. An invalid explicit matrix, an out-of-envelope explicit state, a violated domain condition, or an exhausted iteration or query budget terminates the LP call with FAIL. Pair formulas themselves use the bounded fallbacks just described, so their evaluation and aggregation remain defined even when the failure is not detectable. Thus no unsuccessful execution can request finer precision, larger coefficients, or additional iterations. On the simultaneous spectral-success event all actual-row states satisfy the proved envelopes, so clipping and aborts do not alter the argument. In the cut algorithm, replace FAIL by a fixed legal shore and query its value; this preserves the unconditional query cap and the stated success probability.

15.6 Initialization, cost switch, and final accuracy

Compute a dyadic X^0 within Δ of $\log g(x^0)$, using Lemma 15.9, and put $w^0 = \exp X^0$. Define the artificial cost by the exact public circuit

$$c_i^{\mathrm{art}} = -w_i^0 \phi_i'(x_i^0).$$

Then x^0 is an exact weighted center at $t = 1$, the potential is at most $2Me^{\mu\Delta} < B_0$, and $\|c^{\mathrm{art}}\|_\infty \le 4U$.

Let $V = 20T_0\overline{M}U^2$. Choose a dyadic endpoint

$$\tfrac{1}{2}\min\{1, \Delta/(16kV)\} \le t_- \le \min\{1, \Delta/(16kV)\}.$$

Follow the artificial-cost path from 1 down to t_-, using Proposition 15.6 and (52). At the endpoint perform one additional batch without changing t. Replace c^{art} by c. The centrality increase is at most

$$t_- N_w \left(\frac{c - c^{\mathrm{art}}}{wS} \right) \le t_- V \le \Delta/(16k),$$

which fits inside the margin (50). The potential and primal point are unchanged by the switch. Now follow the true-cost path to a dyadic $t_+ \in [8r/\varepsilon, 16r/\varepsilon]$.

Final centering with frozen weights. Set

$$d_{\mathrm{goal}} = \frac{\varepsilon}{32MU^2}, \qquad J = \max\{0, \lceil \log_2(2\Delta/d_{\mathrm{goal}}) \rceil\},$$

and use throughout the algorithm a local direction tolerance

$$\eta \le \min\left\{ \frac{\Delta}{2^{16}k}, \frac{d_{\mathrm{goal}}}{32}, \frac{\Delta}{2^{10}(J+1)} \right\}. \tag{59}$$

Freeze $t = t_+$ and the current w, and take J repaired Newton steps without weight updates. Lemma 15.4 implies

$$\delta_{j+1} \le \tfrac{1}{2}\delta_j + 4\eta, \qquad \delta_j \le 2^{-j}\Delta + 8\eta,$$

as long as the weight comparison holds. Its total target-weight movement is at most $4\sum_{j<J}(\delta_j + \eta) < 10\Delta$, by (45) and (59). Hence that comparison continues to hold, closing this induction as well. At the end $\delta_J \le d_{\mathrm{goal}}$, and the polynomial slack bound still applies.

Let $\bar{x}$ be the exact center for these frozen final weights. Barrier stationarity and (42) give $c^\top \bar{x} - \mathrm{OPT} \le \sum_i w_i/t_+ \le \varepsilon/2$. Lemma 15.7 and $\|c\|_1 \le MU$ give

$$|c^\top (x^J - \bar{x})| \le 4MU^2 d_{\mathrm{goal}} \le \varepsilon/8.$$

Thus the returned point is ε-suboptimal. It is exactly feasible and strictly interior by the repaired-step construction.

15.7 Closing the induction and charging queries

The precision requirements are now noncircular. The initial point has certified slack. Assuming the invariants at the start of a batch, Lemmas 15.7–15.8 give (54). This licenses the numerical accuracies in Lemma 15.10, the Lewis computation, and the support computation. Proposition 15.6 then proves the invariants for the next state. The separate frozen-weight argument handles the final phase. Because (59), s_*, and q_* are inverse polynomial, one fixed $\mathcal{N}^{-O(1)}$ interface precision suffices. There is no iteration-by-iteration multiplication of unspecified Lipschitz constants.

Each phase traverses a polynomial range of t, so its number of batches is

$$O\left(\frac{kT_0}{\Delta}\log \mathcal{N}\right) = \widetilde{O}(\sqrt{r}).$$

The $J = O(\log \mathcal{N})$ final solves do not increase this bound. Theorem 14.1 and Corollary 14.7 give frozen normal systems a preconditioner with constant accuracy. Refinement against the full dyadic matrix takes $O(\log \mathcal{N})$ Richardson iterations. Leverage estimates need only inverse-polylogarithmic relative accuracy. The Lewis contraction uses $O(\log \mathcal{N})$ such calls. Dyadic matrix products, repair residuals, and the support searches use the dyadic aggregation Lemma 14.8. Each batch therefore costs $r \log^{O(1)} \mathcal{N}$ cut queries.

All operations preserve equality of duplicate hidden coordinates. Normal systems use multiplicity linearly, as separate identical rows. Every hidden output is obtained from public pair formulas, public potentials, and public scalars; the repair changes only explicit rows. Assign each spectral call an inverse-polynomial conditional failure probability and apply the adaptive union bound over a deterministic cap on the calls. On the simultaneous success event the proof applies to the entire run. This completes Theorem 15.1.

Proof and formalization boundary. This argument proves a specified modified execution: fixed-radius batches, absolute errors followed by exact repair on explicit rows, a contractive Lewis restart, certified mixed-ball support optimization, and final frozen-weight centering. It incorporates the regularized-sensitivity and norm-transfer corrections above and the changed polynomial envelopes. It does not validate the old relative-error transfer or the old regularizer. Theorem 15.1 is the interface used by the terminal reduction. The calibrated query-program endpoints and their explicit logarithmic envelopes are described in Section 17. That formal scope is distinct from a literal verification of every numerical constant or general LP instance covered by this section.

16 The accelerated energy inequality

We provide the algebra behind Lemma 7.3, with the same notation as (23). It is useful to keep the error term exactly, rather than bound it separately before the cancellation. Let x, z, v, g, x^+, z^+ be one iteration, let $g = \nabla \mathcal{F}(v) + \eta$, and put $\bar{z} = (1-\alpha)z + \alpha v$. Since $L\alpha^2 = 1$, smoothness gives

$$\mathcal{F}(x^+) \le \mathcal{F}(v) - \frac{\|g\|_2^2}{2L} + \frac{\langle \eta, g\rangle}{L}. \tag{60}$$

Apply 1-strong convexity at v to x and r_*, multiply respectively by $1-\alpha$ and α, and add. The identity $x - v = -\alpha(z - v)$ gives

$$(1-\alpha)(x-v) + \alpha(r_* - v) = \alpha(r_* - \bar{z}).$$

Rearranging the strong-convexity inequality therefore yields

$$\begin{aligned}\mathcal{F}(v) - \mathcal{F}(r_*) \le (1-\alpha)(\mathcal{F}(x) - \mathcal{F}(r_*)) + \alpha\langle \nabla\mathcal{F}(v), \bar{z} - r_*\rangle \\ - \tfrac{1}{2}(1-\alpha)\|x-v\|_2^2 - \tfrac{\alpha}{2}\|v - r_*\|_2^2. \end{aligned} \tag{61}$$

The squared-norm expansion is

$$\begin{aligned}\tfrac{1}{2}\|z^+ - r_*\|_2^2 &= \tfrac{1}{2}\|\bar{z} - r_*\|_2^2 - \alpha\langle g, \bar{z} - r_*\rangle + \tfrac{\alpha^2}{2}\|g\|_2^2, \\ \|\bar{z} - r_*\|_2^2 &= (1-\alpha)\|z - r_*\|_2^2 + \alpha\|v - r_*\|_2^2 - \alpha(1-\alpha)\|z - v\|_2^2. \end{aligned} \tag{62}$$

Add (60)–(62). The $\|g\|^2$ terms cancel because $\alpha^2 = 1/L$, and the $\|v - r_*\|^2$ terms cancel as well. The two error terms combine exactly to

$$\langle \eta, g\rangle/L - \alpha\langle \eta, \bar{z} - r_*\rangle = -\alpha\langle \eta, z^+ - r_*\rangle.$$

Writing $E = \mathcal{F}(x) - \mathcal{F}(r_*) + \|z - r_*\|^2/2$ and similarly for E^+, we obtain

$$E^+ \le (1-\alpha)E - \alpha\langle \eta, z^+ - r_*\rangle - \tfrac{1-\alpha}{2}\|x - v\|_2^2 - \tfrac{\alpha(1-\alpha)}{2}\|z - v\|_2^2.$$

Discarding the two nonpositive terms proves the source recurrence. If $X = \sqrt{E^+}$ and $Y = \sqrt{E}$, then $X^2 \le (1-\alpha)Y^2 + \alpha\epsilon_0\sqrt{2}X$. The positive root is at most $\sqrt{1-\alpha}Y + \alpha\epsilon_0\sqrt{2}$, proving the claimed contraction. This computation requires only strong convexity, smoothness, and the absolute gradient error. It never asks for r_* or for an objective-value oracle.

17 Scope, formalization, and provenance

The outputs of both terminal algorithms are original minimizing shores and exact original cut values. The deterministic representation supports a flow computation, but the represented flow need not be feasible on the original hidden arcs. Nor does a successful randomized transcript by itself certify optimality against every graph consistent with that transcript. The results concern finding cuts in simple unit digraphs. Bounded pair multiplicity, integrality, and free offline computation are used explicitly in the proofs.

Deterministic endpoint. The Lean library [Lee26] contains `exists_unitST_query_program` in the namespace `DirectedUnitST.Endpoint`. For $n \ge 2$ and $s \ne t$, it constructs one public deterministic query tree before the hidden simple digraph is chosen. The tree returns an original minimum s–t shore and its exact natural-valued outgoing-cut value. Its budget is

$$\left\lceil 2048 n^{5/3}(\lceil \log_2 n\rceil + 4)^2 \right\rceil,$$

and its cost on every answer oracle is at most $2049 n^{5/3}(\lceil \log_2 n\rceil + 4)^2$. Preparation, discrepancy repair, cut equivalence, and the final terminal solve are composed in the endpoint, rather than supplied as algorithmic hypotheses.

Randomized endpoints. The declarations `directed_terminal_query_theorem` and `directed_global_query_` in `ImplicitLP/PathFollowing/QueryEndpoint.lean` concern the actual calibrated finite query programs. For $0 < \zeta \leq 1$, they guarantee exact original shore/value outputs with probability at least $1 - \zeta$ and unconditional query bounds

$$2^{310} n^{3/2} \log_2^{21}(2n/\zeta) \quad \text{and} \quad 2^{340} n^{7/4} \log_2^{22}(2n/\zeta),$$

respectively. These conservative explicit envelopes establish the stated powers of n; they are not a claim that each optimized polylogarithmic account in the manuscript has been reproduced literally in Lean. Exact local arithmetic is free in these formal query models as well.

Verification. The library's transitive axiom audit, run with `-trust=0`, covers both endpoint modules and allows only `propext`, `Classical.choice`, and `Quot.sound`. The manuscript retains its full mathematical proofs, including the repaired LP analysis. Appendix A maps every numbered result of the paper to its Lean declarations.

17.1 Sources and provenance

The anonymous note *Directed Reachability from Cut Queries in $\widetilde{O}(n^{3/2})$ Queries* [Anon26] predates this work. Yang P. Liu's repository lists it on 12 August 2026 among AI-generated papers [Liu26]. Its opening note credits discovery of the algorithm and main proof to an agentic harness developed by George Z. Li and Janardhan Kulkarni. This is provenance reported by that source, not an authorship assignment to the repository host or harness developers.

That note supplies the fractional-orientation viewpoint, the rank-sensitive implicit LP approach to reachability, the filtered spectral-oracle framework, the cut-algebra, star-search, and bit-aggregation primitives of Section 6, the star-variable penalty, and the threshold search. We build on those ideas. Our randomized development extends them to exact minimum-cut values and shores and supplies a replacement numerical execution. The audit [LPAudit26] identifies a false general solve-error transfer in the note's stability interface; the later audit [LSAudit26] of the Lee–Sidford facts it relies on identifies a false general sensitivity claim in that supporting chain. These defects concern proof steps, and do not refute the reachability theorem's statement. The repaired bounds and full execution are proved here. In particular, we make no claim to the first subquadratic directed-reachability algorithm.

Weighted path following and the Lewis-weight Jacobian estimates come from Lee and Sidford [LS20], and the Lewis-weight contraction from Cohen and Peng [CP15]. The blocking-flow query accounting of Section 9 follows Anand, Saranurak, and Wang [ASW25], with the layer bounds of Karzanov [Kar73] and Even and Tarjan [ET75], and the global reduction follows the small-shore and sampling balance of [CLN+21]. The spectral construction adapts weighted spanners of Baswana and Sen [BS07], bundle sampling of Koutis [Kou14], and matrix Bernstein in Tropp's form [Tro12]. Every other argument the paper uses is proved in full here; it does not depend on unpublished notes.

AI use

The results in this paper were obtained by prompting Codex GPT-6 Astra (OpenAI), which also wrote this preliminary draft. The Lean proofs (Appendix A) were developed by Claude Opus 5.5 (Anthropic) and audited by GPT-6 Astra.

References

[AEG+22] Simon Apers, Yuval Efron, Paweł Gawrychowski, Troy Lee, Sagnik Mukhopadhyay, and Danupon Nanongkai. *Cut Query Algorithms with Star Contraction*. FOCS, 2022. https://arxiv.org/abs/2201.05674.

[Anon26] Anonymous. *Directed Reachability from Cut Queries in $\widetilde{O}(n^{3/2})$ Queries*. AI-generated research note, listed 12 August 2026 in [Liu26]. The PDF credits an agentic harness developed by George Z. Li and Janardhan Kulkarni with discovery of the algorithm and main proof. Snapshot accessed 11 October 2026; its inexact LP interface is repaired here. https://yangpliu.github.io/repo/directed-reachability-cut-queries/paper.pdf.

[ASW25] Aditya Anand, Thatchaphol Saranurak, and Yunfan Wang. *Deterministic Edge Connectivity and Max Flow using Subquadratic Cut Queries*. SODA, 2025. https://arxiv.org/abs/2410.18704.

[BDP26] Aidan Bai, Chenxin Dai, and Richard Peng. *Bit Complexity of LS'15 under Fixed Point*. AI-generated document, listed 1 October 2026 in [Liu26]. Analyzes the linear-programming method of Lee and Sidford (FOCS 2015) under fixed-point arithmetic. https://yangpliu.github.io/repo/ls15-fixed-point/paper.pdf.

[BS07] Surender Baswana and Sandeep Sen. *A Simple and Linear Time Randomized Algorithm for Computing Sparse Spanners in Weighted Graphs*. Random Structures & Algorithms 30(4), 532–563, 2007. https://www.cse.iitd.ac.in/~ssen/journals/randstruc.pdf.

[BTN26] Ben Bals, Matei Tinca, and Yasamin Nazari. *Cut Query Reachability for DAGs with Subquadratic Queries*. Preprint, 29 July 2026. https://arxiv.org/abs/2607.26630.

[CGJS22] Deeparnab Chakrabarty, Andrei Graur, Haotian Jiang, and Aaron Sidford. *Improved Lower Bounds for Submodular Function Minimization*. FOCS, 2022. https://arxiv.org/abs/2207.04342.

[CLN+21] Ruoxu Cen, Jason Li, Danupon Nanongkai, Debmalya Panigrahi, and Thatchaphol Saranurak. *Minimum Cuts in Directed Graphs via $\sqrt{n}$ Max-Flows*. Preprint, 2021; merged with work of Quanrud into *Minimum Cuts in Directed Graphs via Partial Sparsification*, FOCS, 2021. https://arxiv.org/abs/2104.07898.

[CP15] Michael B. Cohen and Richard Peng. *ℓ_p Row Sampling by Lewis Weights*. STOC, 2015. https://arxiv.org/abs/1412.0588.

[Din70] E. A. Dinic. *Algorithm for Solution of a Problem of Maximum Flow in Networks with Power Estimation*. Soviet Mathematics Doklady 11, 1277–1280, 1970.

[ET75] Shimon Even and Robert Endre Tarjan. *Network Flow and Testing Graph Connectivity*. SIAM Journal on Computing 4(4), 507–518, 1975.

[FF56] L. R. Ford, Jr. and D. R. Fulkerson. *Maximal Flow through a Network*. Canadian Journal of Mathematics 8, 399–404, 1956.

[GK00] Vladimir Grebinski and Gregory Kucherov. *Optimal Reconstruction of Graphs under the Additive Model*. Algorithmica 28(1), 104–124, 2000.

[GKMM26] Prantar Ghosh, Sahil Kuchlous, Shravan Mehra, and Sagnik Mukhopadhyay. *The Power of the Score Sequence of a Tournament*. ESA, 2026. https://arxiv.org/abs/2607.15260.

[GPRW20] Andrei Graur, Tristan Pollner, Vidhya Ramaswamy, and S. Matthew Weinberg. *New Query Lower Bounds for Submodular Function Minimization*. ITCS, 2020. https://arxiv.org/abs/1911.06889.

[Hub64] Peter J. Huber. *Robust Estimation of a Location Parameter*. Annals of Mathematical Statistics 35(1), 73–101, 1964.

[Jia21] Haotian Jiang. *Minimizing Convex Functions with Integral Minimizers*. SODA, 2021; journal version *Minimizing Convex Functions with Rational Minimizers*, Journal of the ACM 70(1). https://arxiv.org/abs/2007.01445.

[JNS26] Yonggang Jiang, Danupon Nanongkai, and Pachara Sawettamalya. *Minimum s–t Cuts with Fewer Cut Queries*. SODA, 2026. Version 1, 21 October 2025. https://arxiv.org/abs/2510.18274v1.

[Kar73] A. V. Karzanov. *On Finding Maximum Flows in Networks with Special Structure and Some Applications*. Matematicheskie Voprosy Upravleniya Proizvodstvom 5, 81–94, Moscow State University Press, 1973. In Russian.

[KK26] Yotam Kenneth-Mordoch and Robert Krauthgamer. *Faster All-Pairs Minimum Cut: Bypassing Exact Max-Flow*. STOC, 2026. Version 2, 6 April 2026. https://arxiv.org/abs/2511.10036v2.

[Kou14] Ioannis Koutis. *Simple parallel and distributed algorithms for spectral graph sparsification*. SPAA, 2014. https://arxiv.org/abs/1402.3851.

[KP12] Michael Kapralov and Rina Panigrahy. *Spectral Sparsification via Random Spanners*. ITCS, 2012.

[KPS26] Sanjeev Khanna, Aaron Putterman, and Junkai Song. *Reachability in Directed Acyclic Graphs with Near-Linear Cut Queries*. Preprint, 23 July 2026. https://arxiv.org/abs/2607.21390.

[Lee26] Troy Lee. *Cut Query Complexity in Lean*, version 0.1.0. Lean 4 library, 2026. https://github.com/troyjlee/cut-query-complexity; doi:10.5281/zenodo.23885654. Directed endpoints: `DirectedUnitST/Endpoint/Main.lean` and `ImplicitLP/PathFollowing/QueryEndpoint.lean` under `CutQueryComplexity/`.

[Lew78] D. R. Lewis. *Finite Dimensional Subspaces of L_p*. Studia Mathematica 63(2), 207–212, 1978.

[Liu26] Yang P. Liu. *Repository*. Catalog of AI-generated papers; accessed 11 October 2026. https://yangpliu.github.io/repository.html. Cited for hosting and provenance, not authorship of [Anon26].

[LLSZ21] Troy Lee, Tongyang Li, Miklos Santha, and Shengyu Zhang. *On the Cut Dimension of a Graph*. CCC, 2021. https://arxiv.org/abs/2011.05085.

[LPAudit26] *Audit of the External Implicit Incidence LP Theorem*. Research note, 4 October 2026. https://github.com/troyjlee/cut-query-complexity/blob/main/papers/P4/supporting/implicit-lp-audit.pdf.

[LS20] Yin Tat Lee and Aaron Sidford. *Solving Linear Programs with $\sqrt{\text{rank}}$ Linear System Solves*. Version 2, 2020. https://arxiv.org/abs/1910.08033v2. The regularized-sensitivity claim is replaced by the proof in Section 15; it is not used as stated in this source.

[LSAudit26] *Audit of the Lee–Sidford facts used by the stable implicit LP*, with numerical checks, October 2026. https://github.com/troyjlee/cut-query-complexity/tree/main/docs/lee-sidford-audit. The repaired sensitivity and movement arguments are restated in Section 15.

[LSW15] Yin Tat Lee, Aaron Sidford, and Sam Chiu-wai Wong. *A Faster Cutting Plane Method and its Implications for Combinatorial and Convex Optimization*. FOCS, 2015. https://arxiv.org/abs/1508.04874.

[MN20] Sagnik Mukhopadhyay and Danupon Nanongkai. *Weighted Min-Cut: Sequential, Cut-Query, and Streaming Algorithms*. STOC, 2020. https://arxiv.org/abs/1911.01651.

[Mon21] Geoffrey Ting-Fai Mon. *Query Complexity of Graph Problems in a Directed Cut Oracle Model*. Bachelor's thesis, Princeton University, 2021. http://arks.princeton.edu/ark:/88435/dsp016395wb216.

[Nan24] Danupon Nanongkai. *Cross-Paradigm Graph Algorithms*. Invited talk, ICALP, 2024; LIPIcs 297, 3:1. The open problem "Cut query for reachability" is on the author's page https://sites.google.com/site/dannanongkai/open, accessed 2 November 2026.

[Nes04] Yurii Nesterov. *Introductory Lectures on Convex Optimization: A Basic Course*. Kluwer, 2004.

[Nes83] Yurii Nesterov. *A Method for Solving the Convex Programming Problem with Convergence Rate $O(1/k^2)$*. Soviet Mathematics Doklady 27(2), 372–376, 1983.

[NN94] Yurii Nesterov and Arkadii Nemirovskii. *Interior-Point Polynomial Algorithms in Convex Programming*. SIAM, 1994.

[RSW18] Aviad Rubinstein, Tselil Schramm, and S. Matthew Weinberg. *Computing Exact Minimum Cuts Without Knowing the Graph*. ITCS, 2018. https://arxiv.org/abs/1711.03165.

[SS11] Daniel A. Spielman and Nikhil Srivastava. *Graph Sparsification by Effective Resistances*. SIAM Journal on Computing 40(6), 1913–1926, 2011.

[Tro12] Joel A. Tropp. *User-Friendly Tail Bounds for Sums of Random Matrices*. Foundations of Computational Mathematics 12, 389–434, 2012. Theorem 6.1(ii); preprint revised 15 June 2011. https://tropp.caltech.edu/papers/Tro11-User-Friendly-preprint.pdf.

A Correspondence with the Lean formalization

This appendix maps every numbered result of the paper to the Lean declarations of [Lee26] that state it. Two modules, `DirectedUnitST/Endpoint/Representation.lean` and `ImplicitLP/PathFollowing/GlobalValidity.lean`, import the endpoint files without changing them, and state the representation sentence of Theorem 1.1 and the legality clause of Theorem 1.2 for the endpoint programs. Declaration names are relative to the namespace `CutQueryComplexity`. Every declaration in the table was elaborated against a fresh build of version 0.1.0, and `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound` for each of them.

Graphs are `DiGraph (Fin n)`, with natural arc multiplicities; `G.Simple` allows at most one arc per ordered pair, and `G.out S` is $F(S)$. The capacitated lemmas use `DirectedCertificate.WeightedDiGraph`, with real capacities. A deterministic program is a `DTree` and a randomized one an `RTree`. Both are built from n and the public parameters before the graph, and the query cap in the program's type holds for every answer function, including failing executions. All constants are explicit. The deterministic statements use the integer logarithm $\lceil \log_2 n \rceil$, the randomized endpoints use $\log_2(2n/\zeta)$, and the spectral primitive uses $D = \ln(2Cn/(\varepsilon\zeta))$. Offline arithmetic is exact real arithmetic, as in the paper's model. Section 16 contains no numbered statement.

Every numbered result has a Lean counterpart, and most are stated with the paper's hypotheses. Three clauses are not exported as statements. Only the "if" direction of Corollary 6.2 is a single statement. Corollary 14.7 is proved as two generic matrix inequalities, which are not instantiated at the sparsifier output. In Theorem 15.1, strict interiority of the iterates and the batch count are internal invariants; the fixed-precision claim is formalized only for the dyadic quantities on the query side, since offline arithmetic is exact. The table records smaller differences of form. The Lean terminal endpoint instantiates the path-following solver for the terminal LP directly, rather than through the general Theorem 10.2.

Paper	Lean	Notes

Theorem 1.1	`DirectedUnitST.Endpoint.exists_unitST_query_program`
	`DirectedUnitST.Endpoint.exists_unitST_representation`
	`DirectedUnitST.Blocking.State.Realizes`
		For $n \geq 2$ and $s \neq t$, one tree of budget $\lceil 2048 n^{5/3}(\lceil \log_2 n \rceil + 4)^2 \rceil$, with cost at most $2049 n^{5/3}(\lceil \log_2 n \rceil + 4)^2$, returns a minimum s–t shore S and $F(S)$ on every simple G. A second tree, built from n alone and within the same cost bound, returns a public state. On every simple G this state realizes an integral representation with the cut function of G: every unknown pair carries its pair total in the public direction, and every other pair has its recorded capacity. The first tree runs its terminal stage on this state.

Paper	Lean	Notes

Lemma 6.1	`DirectedCertificate.WeightedDiGraph.two_mul_out`
	`DirectedCertificate.WeightedDiGraph.symStar_eq`
	`DirectedCertificate.WeightedDiGraph.out_eq_sum_outdeg_sub_internal`
	`DiGraph.two_mul_out`
	`DiGraph.pairs_eq_out`
	`DiGraph.out_eq_sum_outdeg`
	`DirectedUnitST.Preparation.out_eq_of_pair_sum_of_capacityDiv`
		(15)–(17), for real capacities and for multigraphs; `pairs_eq_out` gives $c(A,B) = F(A) + F(B) - F(A \cup B)$ for disjoint A, B. The last declaration is the final sentence: equal pair totals and divergences give equal cut functions.

Corollary 6.2	`DirectedUnitST.Preparation.out_eq_of_pair_sum_of_capacityDiv`
	`DirectedCertificate.WeightedDiGraph.sym_w_eq_three_outs`
	`DirectedCertificate.WeightedDiGraph.out_singleton`
	`DirectedCertificate.WeightedDiGraph.out_compl_singleton`
		The "if" direction is stated. The converse is not a single statement; it follows from the stated formulas for c_{uv}, $F(\{v\})$ and $F(V \setminus \{v\})$.

Lemma 6.3	`firstNbrFilter_correct`
	`firstNbr_correct_clog`
	`allRowsUniv_correct`
	`pairAgg_correct`
		For a duplicate-free public list of at most 2^k vertices other than v, the first positive pair in list order after a public filter, with its multiplicity, or its absence, in $3 + 3k$ queries. The row sums and the sum over $u < v$ of a signed dyadic formula with B bit positions take $6Bn$ queries each. Stated for multigraphs.

Lemma 7.1	`DirectedUnitST.SparseBand.selectOffsets_good` `DirectedUnitST.SparseBand.offsetsProg_good`	For $k > 0$, the table chosen from (n, k) satisfies $\lvert\mathcal{A}(r)\rvert \le H$ for every $r \in \mathbb{R}^n$, with H exactly as in (19). The program returning it has query budget 0.

Lemma 7.2	`DirectedUnitST.Acceleration.graphPotential_lower`
	`DirectedUnitST.Acceleration.graphPotential_upper`
	`DirectedUnitST.Acceleration.canonical_laplacian_bound`
	`DirectedUnitST.Acceleration.graphPotential_coercive_lower`
	`DirectedUnitST.Acceleration.graphPotential_zero_le`
	`DirectedUnitST.Acceleration.graphPotential_initial_energy`
	`DirectedUnitST.Acceleration.graphPotential_exists_minimizer`
		Stated for any weighted arc family, with (20) as the definition of the gradient. The lower and upper quadratic models give 1-strong convexity and $(kB + 1)$-smoothness whenever B bounds the weighted Laplacian; $B = 4n$ for pair totals at most 2. Then $f \ge -3m/2$, $f(0) \le m$, and $\mathcal{F}(0) - \mathcal{F}(r) + \lVert r\rVert_2^2/2 \le 5m/2$ for every r, which gives both bounds at r_*. A minimizer exists. Differentiability and uniqueness are not stated.

Lemma 7.3	`DirectedUnitST.Acceleration.queryPotentialProg_gradient`
	`DirectedUnitST.Acceleration.queryPotentialProg_l1`
	`DirectedUnitST.Acceleration.queryPotentialProg_cost_sqrt`
		For simple G, $n \ge 2$, $k \ge 1$ and every offset table: $\lVert\nabla\mathcal{F}(r)\rVert_2^2 \le 2T^2$, (24), and cost at most $60n(\sqrt{nk} + 1)(\lceil\log_2(64n^2)\rceil + 1)^2$, with $T = \lfloor\sqrt{8nk}\rfloor + 1$.

Paper	Lean	Notes

Lemma 7.4	`DirectedUnitST.Repair.signed_flow_rounding`
	`DirectedUnitST.Preparation.publicCorrection_l1`
	`DirectedUnitST.Preparation.rounded_regularized_error`
	`DirectedUnitST.Preparation.potentialRepresentation_rise`
	`DirectedUnitST.Preparation.realizedRounded_rise`
	`DirectedUnitST.Preparation.integralPreparation_supply_le_half_sq`
	`DirectedUnitST.Preparation.PreparationSpec`
		Rounding with floor and ceiling bounds on arcs and divergences is stated for every real flow. The other clauses are stated for the actual preparation output: an ℓ_1 divergence change of at most $2n$, $E \le 8n\sqrt{k}$, (26), an integral, zero-sum, feasible q, and $R \le n^2/2$. `potentialRepresentation_rise` gives the rise $1 + 1/k$ before rounding.

Lemma 8.1	`DirectedUnitST.Repair.potential_height` `DirectedUnitST.Preparation.integralPreparation_height`	The stated hypotheses and (27). For the preparation output, $R \le 4n^{4/3} + 10n\sqrt{k}$.

Theorem 9.1	`DirectedUnitST.Blocking.discrepancyRepairTree`
	`DirectedUnitST.Blocking.discrepancyRepairTree_correct`
	`DirectedUnitST.Blocking.repairProg_flow_integer`
		The budget, in the tree's type, is 0 if $R = 0$ and otherwise $\lceil (2(n+2) + 4(n+2)\min\{R, 6n^{2/3}\} + 7n\sqrt{R})(2\lceil \log_2 n \rceil + 4) \rceil$. Given a public state realizing an integral representation with pair totals at most 2, an integral demand with $R \le n^2$, and a feasible witness, the tree returns a feasible integral flow with divergence q.

Lemma 9.2	`DirectedUnitST.Blocking.phaseProg_cost_distance_value` `DirectedUnitST.Blocking.bfsRun_cost`	For any level function ℓ on the network of $n+2$ nodes, a phase sending value a costs at most $(2(n+2) + (\ell(t) - \ell(s))a)(2\lceil \log_2 n \rceil + 4)$ queries. The layering search is bounded separately.
Lemma 9.3	`DirectedUnitST.Repair.residual_middle_layer_deficit`	The deficit is at most $8\lvert V \setminus \{s,t\}\rvert^2/d^2$ when $\ell(s) = 0$, $\ell(t) = d \ge 4$, residual arcs rise by at most one level, and internal pair totals are at most 2.

Theorem 1.2	`ImplicitLP.PathFollowing.directed_terminal_query_theorem`
	`ImplicitLP.PathFollowing.directed_global_query_theorem`
	`ImplicitLP.PathFollowing.calibratedTerminalSolver_valid`
	`ImplicitLP.PathFollowing.balancedGlobalMinCut_valid`
		For $n \ge 2$ and $0 < \zeta \le 1$, the caps $2^{310} n^{3/2} \log_2^{21}(2n/\zeta)$ and $2^{340} n^{7/4} \log_2^{22}(2n/\zeta)$ hold on every execution, under every answer function. The success probability is at least $1 - \zeta$ on every simple G; take $\zeta = n^{-c}$. On every execution, the terminal program returns an s–t shore and the global program a nonempty proper shore, each with its value $F(S)$.

Theorem 10.1	`Sparsify.spectral_primitive`
	`Sparsify.budget_le`
	`Sparsify.size_le`
	`Sparsify.spectral_primitive_support`
		As for Theorem 14.1, with ε for θ.

Paper	Lean	Notes

Theorem 10.2	`ImplicitLP.GenericQuery.incidenceLP_query_endpoint`
	`ImplicitLP.GenericQuery.polynomial_input_query_bound`
	`ImplicitLP.GenericQuery.PublicHyp`
	`ImplicitLP.GenericQuery.HonestInput`
	`ImplicitLP.GenericQuery.Solved`
		The general incidence LP: hidden rows of multiplicity at most C, at most $C_{\exp}n$ explicit dyadic rows, squared row norms in $[a_0^2, 2]$, dyadic data bounded by U, and a public start of slack at least $1/U$. With probability at least $1-\zeta$, the output is given by pair formulas, is feasible in the closed box, and is within ε of every feasible point. The cap $2^{330}(C+C_{\exp})n^{3/2}\ln^{21}(2N)$, where $N = 2+n+M+U+\varepsilon^{-1}+a_0^{-1}+\zeta^{-1}$, holds on every execution. For polynomial N it is $O(n^{3/2}\log^{21}(2n))$.

Corollary 10.3	`ImplicitLP.PathFollowing.directed_terminal_query_theorem` `DirectedMincut.terminal_solves`	The terminal endpoint above. `terminal_solves` converts any LP solver with the accuracy required by Lemma 12.1 into an exact terminal mincut algorithm with the same error.

Theorem 10.4	`DirectedMincut.global_solves`
	`DirectedMincut.TermSpec`
	`ImplicitLP.PathFollowing.publicShoreSize`
		Generic in the terminal program. It is assumed to have a worst-case cap q, to return legal shores with their values on every execution, and to be exact with probability $1-\zeta$. For $n \le 2^k$, the error is $(1-h/n)^K+\zeta$ and the cap is $2n+2(2nh+n)(3+3k)+1+2Kq$. The global endpoint uses $h = \min\{n-1, \lceil n^{3/4}\rceil\}$.

Lemma 11.1	`DirectedMincut.obj_ge`
	`DirectedMincut.near_opt`
	`DirectedMincut.exists_feasible_obj_eq`
		The optimum $-\lambda$ is attained for simple G. Near-optimal points satisfy $R \le 2\varepsilon$ and $\lvert q/2-\lambda\rvert \le \varepsilon$. Lean's LP points have one coordinate per vertex pair, shared by its copies.

Lemma 12.1	`DirectedMincut.extraction`
	`DirectedMincut.extract`
	`DirectedMincut.extract_correct`
		For any $\tau > 0$ with $8n^2\tau \le 1$ and $\varepsilon \le \tau/16$, and any estimate within $\tau/4$ on actual pairs, every set containing s that is closed and reachable in J excludes t and is a minimum cut. `extract` uses $2n(3+3k)+1$ queries for $n \le 2^k$, including the value query.

Lemma 13.1	`DirectedMincut.smallShore` `DirectedMincut.smallShore_correct`	The cap is $2n+2(2nh+n)(3+3k)+1$ for $n \le 2^k$. The output is a nonempty proper shore with its value, at most the minimum degree, and at most $F(S)$ whenever $\min\{\lvert S\rvert, \lvert V\setminus S\rvert\} \le h$.

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Theorem 14.1	`Sparsify.spectral_primitive`
	`Sparsify.sparsify`
	`Sparsify.budget_le`
	`Sparsify.size_le`
	`Sparsify.spectral_primitive_support`
		For $n \geq 2$, $C \geq 1$, multiplicities at most C, $w \geq 0$, and $0 < \varepsilon, \zeta \leq 1/2$, the failure probability is at most ζ for the event: (33) holds as an energy inequality for every $x \in \mathbb{R}^n$, and all output pairs have $c_e, w_e > 0$. The cap in the type of `sparsify` is at most $2 \cdot 10^{10} C\varepsilon^{-2} n D^8$, and every output has at most $2 \cdot 10^7 C\varepsilon^{-2} n D^6$ pairs, independently of w. The conditional clause is not stated separately.

Lemma 14.2	`firstNbrFilter_correct` `firstNbr_correct_clog`	As for Lemma 6.3: $3 + 3k$ queries for at most 2^k eligible endpoints.
Lemma 14.3	`Spanner.trial_stretch` `Spanner.trialQ_correct`	For every coin sequence and every $k \geq 1$, the completed trial leaves no residual pair and joins each pair ab of E^P by a path of at most $2k - 1$ selected pairs, each no longer than ab. `trialQ_correct` identifies the query program's output with the trial.
Lemma 14.4	`Spanner.amp_spanner`	With $K = \lceil \log_2 n \rceil$: the failure probability is at most 2^{-J}, and the cap in the type is $6Jn(K+1)(3+3K)$. Every output lies in E^P, has at most $4n(K+1)$ pairs, and has stretch $2K+1$. Taking $J = \lceil \log_2(1/\beta) \rceil$ gives the lemma.
Lemma 14.5	`Spectral.pair_dominated` `Spanner.bundle_dominated`	Proved in the equivalent form $\omega_e(x_u - x_v)^2 \leq (C\alpha/t)\, x^\top L' x$ for every x; the pseudoinverse is not used. In `bundle_dominated`, $\alpha = 2k - 1$.
Lemma 14.6	`Spectral.one_round`	If every sampled pair satisfies the domination bound of Lemma 14.5 with constant λ, the failure probability is at most $2n \exp(-\delta^2/(8\lambda))$, for energies of every x; with $\lambda = C\alpha/t$ this is (37). The bundle is a parameter of the statement. Matrix Bernstein is proved in the vendored `SLT/RMT/MatBern.lean`.
Corollary 14.7	`Spectral.psd_add_lower` `Spectral.psd_add_upper`	The two sides, as generic matrix inequalities for any positive semidefinite H. They are not instantiated at the sparsifier output, and the positive-definiteness clause is not stated.
Lemma 14.8	`allRowsUniv_correct` `pairAgg_correct`	$6Bn$ queries each, for signed-digit dyadic formulas on B public bit positions.

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Theorem 15.1	`ImplicitLP.GenericQuery.incidenceLP_query_endpoint`
	`ImplicitLP.GenericQuery.batchGood_invariant`
	`ImplicitLP.GenericQuery.calibratedDownSteps_le`
		The query conclusion is as for Theorem 10.2, with $r = n$. Strict interiority of the iterates is an internal invariant, and the number of downward path steps is at most $2^{74}\sqrt{n}$ times a sixth power of a logarithm. The precision claim is formalized only for the dyadic quantities on the query side.

Lemma 15.2	`ImplicitLP.sens_regLewis` `ImplicitLP.sens_note`	The stated hypotheses and bound. `sens_note` gives 16 for $p = 1 - 1/L_0$, $\log_2(2M) \le L_0 - 1$, and $c \le 1/(2M)$.

Lemma 15.3	`ImplicitLP.proj_mixed_bound`
	`ImplicitLP.projection_contract`
	`ImplicitLP.eq_projection_note`
		The hypothesis is the diagonal leverage bound used in the proof, in place of the ℓ_∞ comparison. `eq_projection_note` gives the factor $1 + 1/(16k)$ when $\frac{4}{5}g \le w \le \frac{5}{4}g$.

Lemma 15.4	`ImplicitLP.lem_newton` `ImplicitLP.eq_newton`	The weight comparison enters as the bound $N_w(Sh) \le \gamma\delta$ with $\gamma \le 17/16$, which `eq_projection_note` supplies. If $\gamma\delta + \eta \le 1/10$, the conclusion is as stated.
Lemma 15.5	`ImplicitLP.lem_chase` `ImplicitLP.chase_invariant`	$\mathcal{U}$ need only lie between the two cubes and be closed under decreasing coordinate magnitudes; convexity is not used. The relations $\mu = \xi/(12\Delta)$, $\xi \le 1/8$ and $C_* \le \Delta$ are explicit hypotheses. `chase_invariant` is the B_0 clause.
Proposition 15.6	`ImplicitLP.prop_batch` `ImplicitLP.GenericQuery.batchGood_invariant`	With $k = 2L_0$ and the parameter bounds as explicit hypotheses ($\Delta \le k^{-2}2^{-16}$, $\eta \le \Delta/(2^{16}k)$, …): exact feasibility, the potential invariant, and (50). `batchGood_invariant` is the form used by the query program.
Lemma 15.7	`ImplicitLP.lem_center`	Assumes $\delta_t(x, w) \le 1/1000$ in place of $\delta_t(x, w) \le \Delta$, and U at least every box width. Then (53) holds for every exact minimizer $\bar{x}$.
Lemma 15.8	`ImplicitLP.lem_slack`	The stated bound, with any $D \ge \lvert t\rvert \, \lVert c_*\rVert_\infty MU + \sum_j w_j$ in the denominator.

Lemma 15.9	`ImplicitLP.lewisMap_contracting`
	`ImplicitLP.isLewis_iff_fixed`
	`ImplicitLP.lewis_exists_unique`
	`ImplicitLP.inexact_iter`
		For $0 < p < 2$, in logarithmic coordinates: contraction factor $1 - p/2$, fixed points equal to the Lewis weights, and the recurrence in the solved form $E_j \le (1 - p/2)^j E_0 + 2\zeta/p$.

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Lemma 15.10	`ImplicitLP.lem_numeric_abs_env` `ImplicitLP.lem_numeric_query` `ImplicitLP.GenericQuery.newtonStep_numeric`	An explicit polynomial, under explicit envelope hypotheses and precision $\beta \le 1/2$. `lem_numeric_query` uses the Richardson solve with the sparsifier preconditioner, and `newtonStep_numeric` is the form used by the query program.
Lemma 15.11	`ImplicitLP.GenericQuery.supportPub_public` `ImplicitLP.GenericQuery.publicSupport_budget_le`	The output is dyadic, sign-aligned, in $\{N_w \le C_*\}$, and σ-optimal; the polynomial bounds are explicit public parameters. It uses at most $(2N+2)(m+3)\cdot 6Bn$ cut queries, which is at most $2^{34}n$ times a cubed logarithm under the calibrated envelope.